

# Bounded Ratios and Optimal Bounding Constants for Ternary Lorentzian Polynomials

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## Abstract

We study bounded ratios and optimal bounding constants among the normalized coefficients of ternary Lorentzian polynomials. For every fixed  $M$ -convex support and in arbitrary degree, we give an explicit presentation of the bounded-ratio cone in terms of quadratic Hessian slices. We further show that the same local-to-global principle holds for quaternary cubics but fails for full-support quaternary quartics. We then express the optimal bounding constants through a variational formula combining local support functions with linear compatibility constraints between slices. For full support, we determine all compatibility relations in arbitrary degree; in degree three, this yields explicit optimal constants for every two-generator section. Finally, we compare the resulting Lorentzian bounds with those for volume polynomials and rank-three matroid basis profiles.

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## 1. Introduction

Brändén and Huh introduced Lorentzian polynomials as a polynomial counterpart of the Hodge–Riemann relations [4]. The class contains volume polynomials of convex bodies and basis-generating polynomials of matroids, and it provides a common source for log-concavity phenomena in geometry and combinatorics. A homogeneous polynomial with nonnegative coefficients is Lorentzian precisely when its support is  $M$ -convex and the Hessian of every quadratic derivative has at most one positive eigenvalue [4, Theorem 2.25].

We study explicit local presentations and sharp constants for the multiplicative coefficient inequalities forced by this local spectral condition. Write a degree- $d$  polynomial in factorial normalization as

$$f(\mathbf{x}) = \sum_{\alpha \in S} c_{\alpha} \frac{\mathbf{x}^{\alpha}}{\alpha!}, \quad c_{\alpha} > 0,$$

where  $S \subseteq \Delta_3^d$  is  $M$ -convex. For  $w = (w_{\alpha})_{\alpha \in S} \in \mathbb{R}^S$ , set

$$R_w(f) = \prod_{\alpha \in S} c_{\alpha}^{w_{\alpha}}, \quad B_S(w) = \sup_{f \in \mathcal{L}_S^+} R_w(f).$$

We call  $w$  a *bounded ratio* when  $B_S(w)$  is finite. The problem has two parts: one must first determine which exponents are bounded and then find their sharp constants.

The quadratic case identifies the local building blocks. Huang, Huh, Soskin, and Wang proved that the bounded-ratio cone of a positive  $3 \times 3$  Lorentzian matrix  $P = (p_{ij})$  is generated by the three triangular ratios

$$\frac{p_{11}p_{23}}{p_{12}p_{13}}, \quad \frac{p_{22}p_{13}}{p_{12}p_{23}}, \quad \frac{p_{33}p_{12}}{p_{13}p_{23}}$$

[5, Theorem B and Example 1.4]. Pulling these inequalities back through the quadratic derivatives of  $f$  produces bounded ratios among its coefficients. We refer to them as the local ratios.

For  $\beta \in \Delta_3^{d-2}$ , let

$$S_\beta := \{e_i + e_j : \beta + e_i + e_j \in S\} \subseteq \Delta_3^2$$

be the support of the corresponding quadratic Hessian slice. The quadratic analysis gives a finite set of bounded-ratio generators for each nonempty  $S_\beta$ ; extending these generators to the global coefficient space gives a collection  $\mathcal{G}(S) \subseteq \mathbb{R}^S$ . The following is Theorem 3.6.

*Theorem A* (Local-to-global bounded ratios). For every  $M$ -convex support  $S \subseteq \Delta_3^d$ ,

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}(S)).$$

*Concurrent work.* After completing this manuscript and while awaiting comments on the draft, Baldi and Kummer [2] posted a closely related preprint. They prove a general tropical characterization of bounded ratios on semialgebraic sets and, in the Lorentzian setting, identify the bounded-ratio cone for a fixed  $M$ -convex support with the dual cone of  $M$ -convex functions. In the ternary setting, combined with the local exchange characterization of  $M$ -convex functions, their result gives an alternative route to the characterization of the bounded-ratio cone in Theorem 3.6. The results of the present paper were obtained independently. Our main additional directions are the explicit quadratic-slice presentation and its geometry, the determination of optimal bounding constants through compatibility between slices, and the persistence and failure of the local-to-global presentation in four variables.

For full support, the local generators are the elementary rhombus ratios

$$R_{\beta,i}(f) = \frac{c_{\beta+2e_i}c_{\beta+e_j+e_k}}{c_{\beta+e_i+e_j}c_{\beta+e_i+e_k}}, \quad \{i, j, k\} = \{1, 2, 3\}.$$

They span the exposed extreme rays of the bounded-ratio cone, and there are exactly  $3\binom{d}{2}$  such rays (Theorem 3.14).

We next ask whether the local-to-global presentation persists beyond three variables. For an  $M$ -convex support  $S \subseteq \Delta_4^d$ , let  $\mathcal{G}_4(S)$  denote the collection obtained by extending the bounded-ratio directions of its quadratic Hessian slices.

*Theorem B* (Four-variable boundary). For every nonempty  $M$ -convex support  $S \subseteq \Delta_4^3$ ,

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}_4(S)).$$

In contrast, for full support  $S = \Delta_4^4$ ,

$$\text{Cone}(\mathcal{G}_4(S)) \subsetneq \text{BR}(\mathcal{L}_S^+).$$

More precisely,

$$R(f) := \frac{c_{1030}c_{3010}c_{0202}}{c_{2020}c_{1012}c_{1210}}$$

satisfies  $R(f) \leq 64$  for every  $f \in \mathcal{L}_S^+$ , but its exponent vector does not belong to  $\text{Cone}(\mathcal{G}_4(S))$ .

These are Theorems 3.9 and 3.10, respectively. Thus the all-degree local-to-global phenomenon is genuinely special to the ternary setting.

The generating cone does not determine the sharp constants. A coefficient of  $f$  can occur in several quadratic slices, so those slices cannot in general be optimized independently. For a quadratic support  $T$ , let  $h_T$  denote the support function of its feasible region of local logarithmic ratios. Let  $L_S^\perp$  denote the space of changes in local weights that preserve the resulting global coefficient ratio. For a nonnegative collection of local weights  $\lambda = (\lambda_\beta)_\beta$ , let  $H_S(\lambda)$  denote the logarithm of the corresponding sharp global bounding constant.

*Theorem C* (Optimal bounding constants). For every  $M$ -convex support  $S \subseteq \Delta_3^d$  and every nonnegative collection of local weights  $\lambda$ ,

$$H_S(\lambda) = \inf_{\eta \in L_S^\perp} \sum_{\beta \in \Delta_3^{d-2}} h_{S_\beta}(\lambda_\beta + \eta_\beta).$$

This is Theorem 4.6. The formula separates the nonlinear optimization within each quadratic slice from the linear compatibility between different slices.

For full support these compatibility conditions admit a completely explicit description. Write

$$r_{\beta,i} := \log R_{\beta,i}, \quad \beta \in \Delta_3^{d-2}, \quad i \in [3].$$

*Theorem D* (Explicit compatibility). A collection

$$r = (r_{\beta,i})_{\beta \in \Delta_3^{d-2}, i \in [3]}$$

arises from a global coefficient-log array if and only if, for every  $\gamma \in \Delta_3^{d-3}$ ,

$$r_{\gamma+e_3,1} - r_{\gamma+e_2,1} + r_{\gamma+e_1,3} - r_{\gamma+e_2,3} = 0$$

and

$$r_{\gamma+e_3,2} - r_{\gamma+e_1,2} + r_{\gamma+e_2,3} - r_{\gamma+e_1,3} = 0.$$

These  $(d-1)(d-2)$  relations are linearly independent.

This is Theorem 5.3. In particular, the abstract compatibility constraint in Theorem 4.6 becomes completely explicit for full support in every degree.

In degree three, only two compatibility parameters remain. The resulting two-variable formula (Theorem 6.2), together with explicit degenerations, determines the sharp constant on every cone generated by two local ratios (Theorem 6.3).

Finally, we compare these universal bounds with two structured classes. Volume polynomials attain one Lorentzian optimum, whereas another functional has a strictly smaller mixed-volume optimum. Three-color basis profiles of rank-three matroids satisfy a still stronger inequality by the Rayleigh property. Thus geometric and combinatorial realizability can impose restrictions that Lorentzianity alone does not detect.

Section 2 introduces the notation. Section 3 determines the bounded-ratio cone and its geometry. Section 4 derives the variational formula for sharp constants, and Section 5 makes its compatibility constraints explicit. Section 6 treats the full-support cubic case. Section 7 gives the geometric and combinatorial applications. Appendix A studies the local-to-global principle in four variables, proving it for cubics and giving a full-support quartic counterexample. Appendix B generalizes the transportation construction behind this counterexample to finite median graphs in arbitrary dimension and degree.

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## 2. Preliminaries

Throughout, we write  $\mathbb{N} := \mathbb{Z}_{\geq 0}$ . For  $\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{N}^n$ , set  $|\alpha| := \alpha_1 + \dots + \alpha_n$ . For integers  $n \geq 1$  and  $d \geq 0$ , define

$$[n] := \{1, \dots, n\}, \quad \Delta_n^d := \{\alpha \in \mathbb{N}^n : |\alpha| = d\},$$

and let  $e_1, \dots, e_n$  be the standard basis of  $\mathbb{R}^n$ . For  $\alpha \in \mathbb{N}^n$ , write

$$\alpha! := \alpha_1! \cdots \alpha_n!, \quad \mathbf{x}^\alpha := x_1^{\alpha_1} \cdots x_n^{\alpha_n}, \quad \partial^\alpha := \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n},$$

where  $\partial_i = \partial/\partial x_i$ . Every homogeneous polynomial in this paper is written in factorial normalization:

$$f(\mathbf{x}) = \sum_{\alpha \in S} c_\alpha \frac{\mathbf{x}^\alpha}{\alpha!}, \quad S \subseteq \Delta_n^d, \quad c_\alpha > 0.$$

*Definition 2.1* ( $M$ -convex set). A nonempty subset  $J \subseteq \Delta_n^d$  is  $M$ -convex if, for every  $\alpha, \beta \in J$  and every  $i$  with  $\alpha_i > \beta_i$ , there exists  $j$  with  $\alpha_j < \beta_j$  such that

$$\alpha - e_i + e_j \in J, \quad \beta - e_j + e_i \in J.$$

For quadratic slice supports we also allow the empty set by convention.

This symmetric exchange condition is a standard characterization of  $M$ -convexity. See [7] and [4, Section 2.1].

*Definition 2.2* (Lorentzian matrix). A real symmetric matrix is called *Lorentzian* if it has at most one positive eigenvalue.

The convention in [5] also requires nonnegative entries. All Hessian slices considered here have nonnegative entries, so the two conventions agree in our setting.

For  $d \geq 2$  and  $\beta \in \Delta_n^{d-2}$ , the *quadratic Hessian slice* of  $f$  at  $\beta$  is

$$H_\beta(f) := (c_{\beta+e_i+e_j})_{1 \leq i, j \leq n},$$

where  $c_\gamma := 0$  for  $\gamma \notin S$ . By factorial normalization,  $H_\beta(f)$  is exactly the Hessian matrix of the quadratic polynomial  $\partial^\beta f$ .

By [4, Definition 2.6 and Theorem 2.25], a degree- $d$  homogeneous polynomial with nonnegative coefficients is Lorentzian if and only if its support is  $M$ -convex and every  $H_\beta(f)$  is Lorentzian. For an  $M$ -convex set  $S \subseteq \Delta_n^d$ , write  $\mathcal{L}_S^+ := \{f : f \text{ is Lorentzian and } \text{supp}(f) = S\}$ .

*Definition 2.3* (Bounded-ratio cone). Let  $\mathcal{X}$  be a nonempty class of degree- $d$  Lorentzian polynomials with common support  $S$ . For  $w = (w_\alpha)_{\alpha \in S} \in \mathbb{R}^S$ , set  $R_w(f) := \prod_{\alpha \in S} c_\alpha^{w_\alpha}$ . The *bounded-ratio cone* of  $\mathcal{X}$  is

$$\text{BR}(\mathcal{X}) := \{w \in \mathbb{R}^S : \sup_{f \in \mathcal{X}} R_w(f) < \infty\}.$$

For  $w \in \text{BR}(\mathcal{X})$ , its *optimal bounding constant* is  $B_{\mathcal{X}}(w) := \sup_{f \in \mathcal{X}} R_w(f)$ .

Because  $R_{au+bv}(f) = R_u(f)^a R_v(f)^b$  for  $a, b \geq 0$ , the set  $\text{BR}(\mathcal{X})$  is a convex cone. We adopt the convention  $\text{Cone}(\emptyset) = \{0\}$ .

### 3. The bounded-ratio cone

**3.1. Local bounded ratios.** For quadratic supports, we abbreviate  $11 = 2e_1$ ,  $22 = 2e_2$ ,  $33 = 2e_3$ , and  $12 = e_1 + e_2$ ,  $13 = e_1 + e_3$ ,  $23 = e_2 + e_3$ . We identify supports that differ by the natural action of  $S_3$ , and we let the empty support represent the zero quadratic.

LEMMA 3.1 (Quadratic supports). *Up to the natural  $S_3$ -symmetry, the  $M$ -convex subsets of  $\Delta_3^2$  are exactly the following eleven supports:*

$$\emptyset, \{11\}, \{12\}, \{11, 12\}, \{12, 13\}, \{11, 12, 13\}, \{11, 12, 22\}, \{12, 13, 23\}, \\ \{11, 12, 13, 23\}, \{11, 12, 13, 22, 23\}, \{11, 12, 13, 22, 23, 33\}.$$

*Proof.* The symmetric exchange axiom yields two forcing rules. If  $ii, jj \in T$ , then  $ij \in T$ . If  $ii, jk \in T$  and  $\{i, j, k\} = \{1, 2, 3\}$ , then  $ij, ik \in T$ .

We classify  $T$  by the number of diagonal points it contains. With no diagonal point, the nonempty possibilities are  $\{12\}$ ,  $\{12, 13\}$ , and  $\{12, 13, 23\}$ . With one diagonal point, say 11, the absence of 23 leaves  $\{11\}$ ,  $\{11, 12\}$ , and  $\{11, 12, 13\}$  up to symmetry. If  $23 \in T$ , the second forcing rule gives the support  $\{11, 12, 13, 23\}$ . With two diagonal points, say 11 and 22, the first rule gives  $12 \in T$ , while the exchange axiom makes 13 and 23 occur together. This produces  $\{11, 12, 22\}$  and  $\{11, 12, 13, 22, 23\}$ . Three diagonal points force full support. Adding the empty set gives the list, and each listed support satisfies the exchange axiom.  $\square$

For a quadratic support  $T \subseteq \Delta_3^2$ , let  $P = (p_{ij})_{1 \leq i, j \leq 3}$  be symmetric, positive on  $T$ , and zero off  $T$ . By factorial normalization,  $P$  is the Hessian of the corresponding quadratic polynomial. For  $w \in \mathbb{R}^T$ , write  $R_w(P) := \prod_{\alpha \in T} p_\alpha^{w_\alpha}$ .

LEMMA 3.2 (Interlacing–determinant criterion). *Let  $A$  be a real symmetric  $3 \times 3$  matrix. Suppose that one of its  $2 \times 2$  principal minors is negative. Then  $A$  is Lorentzian if and only if  $\det A \geq 0$ .*

*Proof.* Let  $\mu_1 > 0 > \mu_2$  be the eigenvalues of the indicated principal submatrix, and let  $\lambda_1 \geq \lambda_2 \geq \lambda_3$  be those of  $A$ . Cauchy interlacing gives

$$\lambda_1 \geq \mu_1 \geq \lambda_2 \geq \mu_2 \geq \lambda_3.$$

Thus  $\lambda_1 > 0 > \lambda_3$ , so  $A$  is Lorentzian exactly when  $\lambda_2 \leq 0$ . Because  $\det A = \lambda_1 \lambda_2 \lambda_3$ , this is equivalent to  $\det A \geq 0$ .  $\square$

On every support containing the required entries, define

$$U := \frac{p_{11}p_{22}}{p_{12}^2}, \quad X := \frac{p_{11}p_{23}}{p_{12}p_{13}}, \quad Y := \frac{p_{22}p_{13}}{p_{12}p_{23}}, \quad Z := \frac{p_{33}p_{12}}{p_{13}p_{23}}.$$

Their exponent vectors in  $\mathbb{R}^T$  are denoted by  $u, x, y, z$ , respectively. Thus  $R_x(P) = X$  whenever  $X$  is defined on  $T$ , and similarly for the other three ratios.

LEMMA 3.3 (Local Lorentzian tests). *Let  $T \subseteq \Delta_3^2$  be a quadratic  $M$ -convex support, and let  $P$  be a symmetric matrix that is positive on  $T$  and zero off  $T$ . Up to a permutation of the indices, the Lorentzianity of  $P$  is characterized as follows.*

(i) If  $T$  is one of

$$\emptyset, \{11\}, \{12\}, \{11, 12\}, \{12, 13\}, \{11, 12, 13\}, \{12, 13, 23\},$$

then  $P$  is Lorentzian.

(ii) If  $T = \{11, 12, 22\}$ , then  $P$  is Lorentzian if and only if  $U \leq 1$ .

(iii) If  $T = \{11, 12, 13, 23\}$ , then  $P$  is Lorentzian if and only if  $X \leq 2$ .

(iv) If  $T = \{11, 12, 13, 22, 23\}$ , then  $P$  is Lorentzian if and only if  $X + Y \leq 2$ .

(v) If  $T = \Delta_3^2$ , then  $P$  is Lorentzian if and only if

$$XY \leq 1, \quad XZ \leq 1, \quad YZ \leq 1,$$

and

$$XYZ + 2 - X - Y - Z \geq 0.$$

*Proof.* The empty support and  $\{11\}$  are immediate. Every other support in (i) has a negative  $2 \times 2$  principal minor. Its determinant vanishes unless  $T = \{12, 13, 23\}$ , for which  $\det P = 2p_{12}p_{13}p_{23} > 0$ . Lemma 3.2 proves (i).

For (ii), the nonzero  $2 \times 2$  block has determinant  $p_{12}^2(U - 1)$ . In (iii) and (iv), respectively,

$$\det P = p_{12}p_{13}p_{23}(2 - X), \quad \det P = p_{12}p_{13}p_{23}(2 - X - Y).$$

Both matrices have a negative principal minor. The three asserted criteria now follow from Lemma 3.2.

Suppose finally that  $T$  has full support. The three  $2 \times 2$  principal minors are nonpositive precisely when  $XY, XZ, YZ \leq 1$ , and

$$\frac{\det P}{p_{12}p_{13}p_{23}} = XYZ + 2 - X - Y - Z.$$

If  $P$  is Lorentzian, interlacing gives the three principal-minor inequalities. Since the positive entries ensure a positive eigenvalue, the other two eigenvalues are nonpositive and  $\det P \geq 0$ .

Conversely, if one principal minor is negative, the determinant inequality and Lemma 3.2 show that  $P$  is Lorentzian. If none is negative, all three vanish. Positivity then makes  $P$  a rank-one positive semidefinite matrix, which is again Lorentzian.  $\square$

The preceding feasible-region description determines the local bounded-ratio cones after passing to logarithmic coordinates.

**COROLLARY 3.4** (Local bounded-ratio cones). *With the cases and notation of Lemma 3.3, the cone  $\text{BR}(\mathcal{L}_T^+)$  is, respectively,*

$$\{0\}, \text{Cone}\{u\}, \text{Cone}\{x\}, \text{Cone}\{x, y\}, \text{Cone}\{x, y, z\}.$$

*Proof.* In case (i), the logarithmic Lorentzian locus is  $\mathbb{R}^T$ , on which only the zero linear functional is bounded above. The inequalities in Lemma 3.3



bound the listed generators in cases (ii)–(iv). For full support, the inequalities  $XZ \leq 1$  and  $XYZ + 2 - X - Y - Z \geq 0$  imply

$$0 \leq XYZ + 2 - X - Y - Z \leq 2 - X - Z.$$

Hence  $X + Z \leq 2$ , and cyclic symmetry gives  $X + Y, Y + Z \leq 2$ . The cone displayed in the statement is therefore contained in  $\text{BR}(\mathcal{L}_T^+)$ .

For the reverse inclusion, consider a support from cases (ii)–(v), and let  $G$  be its listed set of generators. If  $w \notin \text{Cone}(G)$ , the separating hyperplane theorem gives  $q \in \mathbb{R}^T$  such that  $\langle g, q \rangle \leq 0$  for every  $g \in G$ , but  $\langle w, q \rangle > 0$ . For  $t \geq 0$ , set  $p_\alpha(t) = e^{tq_\alpha}$  on  $T$  and set all other entries to zero. Every generator ratio of  $P(t)$  is at most 1, so Lemma 3.3 proves Lorentzianity in cases (ii)–(iv). In case (v), the product inequalities hold, and

$$XYZ + 2 - X - Y - Z = (1 - X)(1 - Y) + (1 - Z)(1 - XY) \geq 0.$$

This factorization proves Lorentzianity in the remaining case. Meanwhile,  $R_w(P(t)) = e^{t\langle w, q \rangle} \rightarrow \infty$ , so  $w$  is not bounded.  $\square$

For full support, Corollary 3.4 recovers the bounded-ratio description in [5, Theorem B and Example 1.4].

**COROLLARY 3.5** (Unit local generators). *Let  $T \subseteq \Delta_3^2$  be a quadratic  $M$ -convex support, and let  $P$  have positive entries on  $T$  and zero entries off  $T$ . If every local generator ratio appearing in Corollary 3.4 is at most 1, then  $P$  is Lorentzian.*

*Proof.* The local criteria settle every non-full support. Under full support, the hypotheses imply the three product inequalities. They also make both terms in  $XYZ + 2 - X - Y - Z = (1 - X)(1 - Y) + (1 - Z)(1 - XY)$  nonnegative, which supplies the remaining determinant inequality in Lemma 3.3(v).  $\square$

**3.2. Global bounded ratios.** Fix  $d \geq 2$  and an  $M$ -convex support  $S \subseteq \Delta_3^d$ . For  $\beta \in \Delta_3^{d-2}$ , define the quadratic slice support

$$S_\beta := \{e_i + e_j : \beta + e_i + e_j \in S\} \subseteq \Delta_3^2.$$

The exchange axiom restricts to  $S_\beta$ . Hence every slice support is  $M$ -convex and appears in Lemma 3.1.

For each  $\beta \in \Delta_3^{d-2}$  and each local generator  $g = \sum_{\gamma \in S_\beta} g_\gamma \varepsilon_\gamma \in \mathbb{R}^{S_\beta}$  from Corollary 3.4, define its extension to  $\mathbb{R}^S$  by

$$\tilde{g}^\beta := \sum_{\gamma \in S_\beta} g_\gamma \varepsilon_{\beta + \gamma},$$

where  $\varepsilon_\gamma$  and  $\varepsilon_\alpha$  denote the standard basis vectors of  $\mathbb{R}^{S_\beta}$  and  $\mathbb{R}^S$ , respectively. Let

$$\mathcal{G}(S) := \{\tilde{g}^\beta : \beta \in \Delta_3^{d-2}, \quad g \text{ is a local generator for } S_\beta\}.$$

The class  $\mathcal{L}_S^+$  is nonempty. Indeed, the polynomial with  $c_\alpha = 1$  for every  $\alpha \in S$  has all local generator ratios equal to 1. Corollary 3.5 and the Hessian-slice characterization therefore show that this polynomial is Lorentzian.

**THEOREM 3.6** (Local-to-global bounded-ratio theorem). *Let  $S \subseteq \Delta_3^d$  be  $M$ -convex. Then*

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}(S)).$$

*Proof.* If  $f \in \mathcal{L}_S^+$ , every  $H_\beta(f)$  is Lorentzian. The local generators on  $S_\beta$  are bounded by Corollary 3.4. Their extensions therefore generate bounded ratios on  $\mathcal{L}_S^+$ , which proves  $\text{Cone}(\mathcal{G}(S)) \subseteq \text{BR}(\mathcal{L}_S^+)$ .

For the reverse inclusion, let  $w \notin \text{Cone}(\mathcal{G}(S))$ . Since  $\text{Cone}(\mathcal{G}(S))$  is a finitely generated closed convex cone, the separating hyperplane theorem gives  $q = (q_\alpha)_{\alpha \in S} \in \mathbb{R}^S$  such that

$$\langle g, q \rangle \leq 0 \quad \text{for every } g \in \mathcal{G}(S), \quad \langle w, q \rangle > 0.$$

For  $t \geq 0$ , define

$$f_t(\mathbf{x}) := \sum_{\alpha \in S} e^{tq_\alpha} \frac{\mathbf{x}^\alpha}{\alpha!}.$$

For a local generator  $g \in \mathbb{R}^{S_\beta}$ , its extension  $\tilde{g}^\beta$  belongs to  $\mathcal{G}(S)$ , and therefore

$$\log R_g(H_\beta(f_t)) = t \langle \tilde{g}^\beta, q \rangle \leq 0.$$

Every local generator ratio is at most 1. Hence Corollary 3.5 makes every  $H_\beta(f_t)$  Lorentzian, and the Hessian-slice characterization gives  $f_t \in \mathcal{L}_S^+$  for all  $t \geq 0$ . On the other hand,

$$R_w(f_t) = \prod_{\alpha \in S} e^{tq_\alpha w_\alpha} = e^{t \langle w, q \rangle} \longrightarrow \infty \quad (t \rightarrow \infty).$$

Thus  $w \notin \text{BR}(\mathcal{L}_S^+)$ , proving the reverse inclusion.  $\square$

For full support, write  $\mathcal{L}_{3,d}^+ := \mathcal{L}_{\Delta_3^d}^+$ . For  $\beta \in \Delta_3^{d-2}$  and  $\{i, j, k\} = \{1, 2, 3\}$ , define

$$g_{\beta,i} := \varepsilon_{\beta+2e_i} + \varepsilon_{\beta+e_j+e_k} - \varepsilon_{\beta+e_i+e_j} - \varepsilon_{\beta+e_i+e_k} \in \mathbb{R}^{\Delta_3^d}.$$

Its associated elementary rhombus ratio is

$$R_{g_{\beta,i}}(f) = \frac{c_{\beta+2e_i} c_{\beta+e_j+e_k}}{c_{\beta+e_i+e_j} c_{\beta+e_i+e_k}}.$$

**COROLLARY 3.7** (Full support). *For every  $d \geq 2$ ,*

$$\text{BR}(\mathcal{L}_{3,d}^+) = \text{Cone} \left\{ g_{\beta,i} : \beta \in \Delta_3^{d-2}, \quad i \in [3] \right\}.$$

*Proof.* Every slice has support  $\Delta_3^2$ . By Corollary 3.4, its extended local generators are  $g_{\beta,i}$  for  $i \in [3]$ . Theorem 3.6 gives the claim.  $\square$

*Remark 3.8* (Relation to Baldi–Kummer). Baldi and Kummer [2] identify the bounded-ratio cone for Lorentzian polynomials with fixed  $M$ -convex support  $S$  with the dual cone of  $M$ -convex functions on  $S$ . In three variables, the local exchange characterization of  $M$ -convex functions identifies this dual cone with  $\text{Cone}(\mathcal{G}(S))$ . Thus their result gives an alternative route to Theorem 3.6. We retain the direct proof above because the explicit quadratic-slice description is used throughout the subsequent analysis.

It is natural to ask whether the local-to-global phenomenon in Theorem 3.6 persists beyond three variables. We now analyze the quaternary case. For an  $M$ -convex support  $S \subseteq \Delta_4^d$  and  $\beta \in \Delta_4^{d-2}$ , set

$$S_\beta := \{\gamma \in \Delta_4^2 : \beta + \gamma \in S\}.$$

Every nonempty  $S_\beta$  is  $M$ -convex, since the common summand  $\beta$  cancels in the symmetric exchange axiom. If  $v = \sum_{\gamma \in S_\beta} v_\gamma \varepsilon_\gamma \in \mathbb{R}^{S_\beta}$ , write

$$\tilde{v}^\beta := \sum_{\gamma \in S_\beta} v_\gamma \varepsilon_{\beta+\gamma} \in \mathbb{R}^S,$$

and define

$$\mathcal{G}_4(S) := \bigcup_{\substack{\beta \in \Delta_4^{d-2} \\ S_\beta \neq \emptyset}} \{\tilde{v}^\beta : v \in \text{BR}(\mathcal{L}_{S_\beta}^+)\}.$$

Thus  $\text{Cone}(\mathcal{G}_4(S))$  is the cone generated by all bounded ratios arising from quadratic Hessian slices.

**THEOREM 3.9** (Quaternary cubics). *For every nonempty  $M$ -convex  $S \subseteq \Delta_4^3$ ,*

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}_4(S)).$$

*Equivalently, every bounded monomial ratio among the normalized coefficients of a quaternary Lorentzian cubic is generated by bounded ratios arising from its quadratic Hessian slices.*

The new difficulty in four variables is the four-point branching phenomenon in a quadratic slice. In degree three, the rigidity imposed by  $M$ -convexity is sufficient to overcome this obstruction. For a complete proof, see Appendix A.1.

**THEOREM 3.10** (Quaternary quartics). *For  $S = \Delta_4^4$ ,*

$$\text{Cone}(\mathcal{G}_4(S)) \subsetneq \text{BR}(\mathcal{L}_S^+).$$

*More precisely,*

$$(3.1) \quad R(f) := \frac{c_{1030}c_{3010}c_{0202}}{c_{2020}c_{1012}c_{1210}}$$

satisfies  $R(f) \leq 64$  for every  $f \in \mathcal{L}_S^+$ , but its exponent vector does not belong to  $\text{Cone}(\mathcal{G}_4(S))$ . The constant 64 is not claimed to be optimal.

The counterexample is obtained by combining local quadratic inequalities to prove boundedness and constructing a separating functional from a simple transportation cost problem on the four vertices of a square. For a complete proof, see Appendix A.2.

Together, these two results clarify the role of the ternary hypothesis in Theorem 3.6. The local-to-global principle does not fail merely upon passing to four variables, since it still holds for all quaternary cubics; however, its failure already for full-support quaternary quartics shows that the validity of the principle in every degree is a genuinely special feature of the ternary case. In this sense, the ternary setting studied in the remainder of the paper is not merely a low-dimensional restriction, but a setting in which bounded ratios admit an all-degree local description that no longer persists in four variables.

**3.3. Geometry of the bounded-ratio cone.** For a positive full-support symmetric matrix  $P = (p_{ij})$ , retain the ratios  $X, Y, Z$  defined before Lemma 3.3 and write  $(x, y, z) = (\log X, \log Y, \log Z)$ . Let  $\mathcal{C} \subseteq \mathbb{R}^3$  be the set of all such triples arising from positive full-support Lorentzian matrices.

**LEMMA 3.11** (Convexity of the full-support log-feasible set). *The set  $\mathcal{C}$  is closed and convex.*

*Proof.* By Lemma 3.3(v), the coordinatewise exponential image of  $\mathcal{C}$  consists of the positive triples satisfying  $XY, XZ, YZ \leq 1$  and  $XYZ + 2 - X - Y - Z \geq 0$ . Every positive triple occurs as the ratio triple of the matrix with  $p_{12} = p_{13} = p_{23} = 1$  and  $(p_{11}, p_{22}, p_{33}) = (X, Y, Z)$ . These inequalities therefore give a complete description.

Since  $XY \leq 1$ , we have

$$0 \leq XYZ + 2 - X - Y - Z = Z(XY - 1) + 2 - X - Y \leq 2 - X - Y,$$

and hence  $X + Y \leq 2$ . Equality forces  $XY = 1$ , so  $X = Y = 1$  and  $0 < Z \leq 1$ .

On the interior  $X + Y < 2$ , we also have  $XY < 1$ , and the determinant condition becomes

$$Z \leq \frac{2 - X - Y}{1 - XY}.$$

This upper bound also implies  $XZ, YZ \leq 1$ . For example, multiplying the bound by  $X$  reduces the desired inequality to  $-(X - 1)^2 \leq 0$ , and the argument for  $Y$  is identical.

It follows that  $\mathcal{C}$  is the closure of the hypograph  $z \leq \phi(x, y)$  over the convex domain  $e^x + e^y < 2$ , where

$$\phi(x, y) = \log(2 - e^x - e^y) - \log(1 - e^{x+y}).$$

Set  $a = e^x$ ,  $b = e^y$ ,  $A = 2 - a - b$ , and  $B = 1 - ab$ . On the interior,  $A, B > 0$ , and differentiation gives

$$\begin{aligned}\phi_{xx} &= \frac{a(b-1)^2(a^2b+b-2)}{B^2A^2}, \\ \det(\nabla^2\phi) &= \frac{2ab(a-1)^2(b-1)^2}{B^2A^3} \geq 0.\end{aligned}$$

Since  $a+b < 2$  and  $(2-a)(1+a^2) - 2 = -a(a-1)^2 \leq 0$ , we have  $b(1+a^2) < 2$  and  $\phi_{xx} \leq 0$ . Symmetry gives  $\phi_{yy} \leq 0$ . The Hessian is therefore negative semidefinite, so  $\phi$  is concave. Its hypograph is convex, as is its closure  $\mathcal{C}$ . The defining inequalities also show that  $\mathcal{C}$  is closed.  $\square$

**THEOREM 3.12** (Multiplicative convexity). *Let  $d \geq 2$  and  $S \subseteq \Delta_3^d$  be  $M$ -convex. If  $f, h \in \mathcal{L}_S^+$  have normalized coefficients  $(c_\alpha)$  and  $(d_\alpha)$ , then for  $0 \leq t \leq 1$  the polynomial*

$$f_t(\mathbf{x}) := \sum_{\alpha \in S} c_\alpha^t d_\alpha^{1-t} \frac{\mathbf{x}^\alpha}{\alpha!}$$

*belongs to  $\mathcal{L}_S^+$ . Equivalently, the logarithmic coefficient locus*

$$\mathcal{A}_S := \left\{ (\log c_\alpha)_{\alpha \in S} : \sum_{\alpha \in S} c_\alpha \frac{\mathbf{x}^\alpha}{\alpha!} \in \mathcal{L}_S^+ \right\}$$

*is convex.*

*Proof.* Fix a slice and write  $u = (\log p_\gamma)_{\gamma \in S_\beta}$  for its coefficient-log vector. Geometric interpolation is affine in  $u$ , while every local log-ratio is linear in  $u$ . We only need to check that each local feasible set is convex.

The supports with no generators give the entire coordinate space, and the one-generator cases give half-spaces. For the five-point support, the ratio-log region  $e^r + e^s \leq 2$  is convex, so its inverse image under  $u \mapsto (r, s) = (\log X, \log Y)$  is convex. For full support, the same conclusion follows from Lemma 3.11. Thus every slice of  $f_t$  is Lorentzian. The Hessian-slice characterization then yields  $f_t \in \mathcal{L}_S^+$ .  $\square$

For a nonempty convex set  $C \subseteq \mathbb{R}^S$  and a cone  $K \subseteq \mathbb{R}^S$ , write

$$\text{rec}(C) := \{q : a + tq \in C \text{ for every } a \in C \text{ and } t \geq 0\},$$

$$K^\circ := \{q : \langle w, q \rangle \leq 0 \text{ for every } w \in K\}$$

for the recession cone of  $C$  and the polar cone of  $K$ , respectively.

**THEOREM 3.13** (Degeneration directions). *Let  $d \geq 2$ , let  $S \subseteq \Delta_3^d$  be  $M$ -convex, and let  $\mathcal{A}_S$  be the logarithmic coefficient locus from Theorem 3.12. Then*

$$\text{rec}(\mathcal{A}_S) = \text{Cone}(\mathcal{G}(S))^\circ = \left\{ q \in \mathbb{R}^S : \langle g, q \rangle \leq 0 \text{ for every } g \in \mathcal{G}(S) \right\}.$$

Consequently,  $\text{BR}(\mathcal{L}_S^+) = \text{rec}(\mathcal{A}_S)^\circ$ .

*Proof.* Suppose  $\langle g, q \rangle \leq 0$  for every  $g \in \mathcal{G}(S)$ , and fix  $a \in \mathcal{A}_S$ . Along  $a + tq$ , every local generator ratio is nonincreasing.

Every non-full slice remains feasible along this ray. A slice with no generator has no constraint. A single upper bound governs each one-generator case, and  $X + Y \leq 2$  governs the five-point case.

For full support, feasibility is characterized by  $XY, XZ, YZ \leq 1$  and  $F(X, Y, Z) := XYZ + 2 - X - Y - Z \geq 0$ . Coordinatewise decrease preserves the product inequalities. It also preserves the determinant inequality, because  $\partial F / \partial X = YZ - 1 \leq 0$  and the cyclic derivatives are likewise nonpositive on the feasible region.

Every slice remains Lorentzian along the ray. Since the support remains  $S$ , we have  $a + tq \in \mathcal{A}_S$  for all  $t \geq 0$  and hence  $q \in \text{rec}(\mathcal{A}_S)$ .

Conversely, let  $q \in \text{rec}(\mathcal{A}_S)$  and choose  $a \in \mathcal{A}_S$ . Theorem 3.6 bounds every  $g \in \mathcal{G}(S)$ . Its ratio along the ray is  $\exp(\langle g, a \rangle + t\langle g, q \rangle)$ , so boundedness forces  $\langle g, q \rangle \leq 0$ . This proves the formula for the recession cone.

The cone  $\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}(S))$  is finitely generated and therefore closed. The remaining identity follows from the bipolar theorem.  $\square$

**THEOREM 3.14** (Exposed extreme rays). *Let  $d \geq 2$ . For full support  $S = \Delta_3^d$ , every ray  $\mathbb{R}_{\geq 0}g_{\beta,i}$ , where  $\beta \in \Delta_3^{d-2}$  and  $i \in [3]$ , is an exposed extreme ray of  $\text{BR}(\mathcal{L}_{3,d}^+)$ . Consequently, the full-support bounded-ratio cone has exactly  $3\binom{d}{2}$  extreme rays, and the generating set in Corollary 3.7 is minimal.*

*Proof.* Put  $q_\alpha^{(0)} = -\|\alpha\|_2^2$ . Every generator  $g$  has norm 2, and a direct calculation gives  $\langle g, q^{(0)} \rangle = -2$ . Fix  $g_0$  and set  $q = q^{(0)} + \frac{1}{2}g_0$ . Then  $\langle g_0, q \rangle = -2 + \frac{1}{2}\|g_0\|_2^2 = 0$ .

For any other generator  $g$ , equality cannot hold in Cauchy–Schwarz. Hence  $\langle g, g_0 \rangle < 4$  and  $\langle g, q \rangle = -2 + \frac{1}{2}\langle g, g_0 \rangle < 0$ .

By Corollary 3.7, the functional  $w \mapsto \langle w, q \rangle$  is nonpositive on the bounded-ratio cone and vanishes there precisely on  $\mathbb{R}_{\geq 0}g_0$ . The ray is therefore exposed, and the construction applies to every elementary generator.

To count the rays, note that the two positive support points of  $g_{\beta,i}$  are  $\beta + 2e_i$  and  $\beta + e_j + e_k$ , where  $\{i, j, k\} = \{1, 2, 3\}$ . Their difference, up to sign, is  $2e_i - e_j - e_k$ , which determines  $i$ . The point with the larger  $i$ th coordinate then determines  $\beta + 2e_i$  and hence  $\beta$ . Thus the indexed generators are pairwise distinct.

Because all generators have the same norm, distinct generators span distinct positive rays. Their number is  $3|\Delta_3^{d-2}| = 3\binom{d}{2}$ . Since they generate the cone, no other extreme rays occur.  $\square$

#### 4. Optimal bounding constants

The bounded-ratio cone answers the finiteness question but does not determine the sharp bounds. We now encode each quadratic optimization by a support function and then impose the compatibility between slices.

For a nonempty set  $C \subseteq \mathbb{R}^N$ , write  $\sigma_C(\lambda) := \sup_{x \in C} \langle \lambda, x \rangle$  for its support function.

For a quadratic  $M$ -convex support  $T$ , fix an ordering  $g_{T,1}, \dots, g_{T,m(T)}$  of the generators from Corollary 3.4. We use the order  $i = 1, 2, 3$  for full support. For a Lorentzian matrix  $P$  with support  $T$ , define

$$\begin{aligned} \ell_T(P) &:= (\log R_{g_{T,1}}(P), \dots, \log R_{g_{T,m(T)}}(P)), \\ \mathcal{C}_T &:= \{\ell_T(P) : P \text{ Lorentzian with support } T\}. \end{aligned}$$

When  $m(T) = 0$ , we set  $\mathcal{C}_T = \{0\} \subseteq \mathbb{R}^0$ . The support function  $h_T := \sigma_{\mathcal{C}_T}$  records the logarithm of the local sharp constant:

$$h_T(\lambda) = \log \sup_P \prod_{\ell=1}^{m(T)} R_{g_{T,\ell}}(P)^{\lambda_\ell}.$$

LEMMA 4.1 (Local support functions). *Up to permutation of the indices, the local log-feasible sets and their support functions are as follows.*

- (i) *For the supports with no nontrivial local generators,  $\mathcal{C}_T = \{0\} \subseteq \mathbb{R}^0$  and  $h_T(0) = 0$ .*
- (ii) *If  $T = \{11, 12, 22\}$ , then*

$$\mathcal{C}_T = (-\infty, 0], \quad h_T(a) = \begin{cases} 0, & a \geq 0, \\ +\infty, & a < 0. \end{cases}$$

- (iii) *If  $T = \{11, 12, 13, 23\}$ , then*

$$\mathcal{C}_T = (-\infty, \log 2], \quad h_T(a) = \begin{cases} a \log 2, & a \geq 0, \\ +\infty, & a < 0. \end{cases}$$

- (iv) *If  $T = \{11, 12, 13, 22, 23\}$ , then*

$$\mathcal{C}_T = \{(r, s) \in \mathbb{R}^2 : e^r + e^s \leq 2\}.$$

For  $a, b \geq 0$ ,

$$h_T(a, b) = (a + b) \log 2 + a \log a + b \log b - (a + b) \log(a + b),$$

with the convention  $0 \log 0 = 0$ . If  $a < 0$  or  $b < 0$ , then  $h_T(a, b) = +\infty$ .

- (v) *For full support  $T = \Delta_3^2$ , we have  $\mathcal{C}_T = \mathcal{C}$ , where  $\mathcal{C}$  is the full-support log-feasible set introduced before Lemma 3.11.*

*Proof.* Parts (i)–(iii) are immediate from Lemma 3.3. In (iv), maximizing  $ar + bs$  over  $e^r + e^s \leq 2$  is equivalent to maximizing  $X^a Y^b$  over  $X + Y \leq 2$ . For  $a, b > 0$ , the maximizer is  $(X, Y) = (2a/(a+b), 2b/(a+b))$ , and continuity covers zero exponents. If  $a < 0$ , for example, the objective becomes unbounded by fixing  $s = 0$  and letting  $r \rightarrow -\infty$ . Part (v) is the definition of  $\mathcal{C}$ .  $\square$

*Remark 4.2.* For full support, write  $h := h_{\Delta_3^2}$ . Then

$$h(a, b, c) = \sigma_{\mathcal{C}}(a, b, c) = \log \sup X^a Y^b Z^c,$$

where the supremum ranges over positive full-support Lorentzian matrices. Theorem C of [5] gives  $\exp h(a, b, c)$  explicitly when  $a, b, c \geq 0$  and  $a + b + c = 1$ . Positive homogeneity extends the formula to the entire nonnegative orthant: if  $s = a + b + c > 0$ , then  $h(a, b, c) = sh(a/s, b/s, c/s)$ . Together with  $h(0, 0, 0) = 0$ , this determines  $h$  on  $\mathbb{R}_{\geq 0}^3$ . Corollary 3.4 gives  $h(a, b, c) = +\infty$  whenever one coordinate is negative.

We use the following standard duality statement to impose compatibility. For a nonempty convex set  $C \subseteq \mathbb{R}^N$ , let  $\text{ri}(C)$  denote its relative interior.

LEMMA 4.3 (Support functions under linear constraints). *Let  $C \subseteq \mathbb{R}^N$  be a nonempty closed convex set, and let  $L \subseteq \mathbb{R}^N$  be a linear subspace. If  $\text{ri}(C) \cap L \neq \emptyset$ , then, for every  $\lambda \in \mathbb{R}^N$ ,*

$$\sigma_{C \cap L}(\lambda) = \inf_{\eta \in L^\perp} \sigma_C(\lambda + \eta).$$

*Proof.* Let  $\delta_C$  and  $\delta_L$  denote the convex indicator functions of  $C$  and  $L$ . The relative-interior hypothesis allows us to apply [8, Theorem 16.4] to  $\delta_{C \cap L} = \delta_C + \delta_L$ , which gives

$$\sigma_{C \cap L}(\lambda) = \inf_{\eta \in \mathbb{R}^N} (\sigma_C(\lambda + \eta) + \delta_L^*(-\eta)).$$

The conjugate  $\delta_L^*(-\eta)$  vanishes on  $L^\perp$  and is infinite elsewhere, which proves the formula.  $\square$

Fix  $d \geq 2$  and an  $M$ -convex support  $S \subseteq \Delta_3^d$ . To assemble the ratio coordinates of its slices, retain the local orderings and set

$$I(S) := \{(\beta, \ell) : \beta \in \Delta_3^{d-2}, 1 \leq \ell \leq m(S_\beta)\}, \quad \mathbb{R}^{I(S)} \cong \prod_{\beta} \mathbb{R}^{m(S_\beta)}.$$

Define  $\rho_S : \mathbb{R}^S \rightarrow \mathbb{R}^{I(S)}$  by

$$(\rho_S(a))_{\beta, \ell} := \sum_{\gamma \in S_\beta} (g_{S_\beta, \ell})_\gamma a_{\beta + \gamma} = \langle \tilde{g}_{S_\beta, \ell}^\beta, a \rangle.$$

Let  $C_S := \prod_{\beta \in \Delta_3^{d-2}} \mathcal{C}_{S_\beta}$  and  $L_S := \text{im}(\rho_S) \subseteq \mathbb{R}^{I(S)}$ . The product  $C_S$  treats the slices independently, whereas  $L_S$  imposes their common coefficient array.



With respect to the standard inner products, the adjoint map is

$$\rho_S^*(\mu) = \sum_{(\beta, \ell) \in I(S)} \mu_{\beta, \ell} \tilde{g}_{S_{\beta, \ell}}^\beta, \quad \mu \in \mathbb{R}^{I(S)}.$$

Thus  $L_S^\perp = \ker(\rho_S^*)$ . For  $f \in \mathcal{L}_S^+$  with  $a_\alpha = \log c_\alpha$ ,

$$(\rho_S(a))_{\beta, \ell} = \log R_{\beta, \ell}(f), \quad R_{\beta, \ell}(f) := R_{g_{S_{\beta, \ell}}}(H_\beta(f)).$$

LEMMA 4.4 (A strictly feasible compatible point). *For  $a^\circ := -\|\alpha\|_2^2$  one has  $\rho_S(a^\circ) \in \text{ri}(C_S)$ . Consequently,  $L_S \cap \text{ri}(C_S) \neq \emptyset$ .*

*Proof.* On  $a^\circ$ , every  $U$ -type generator has value  $-4$ , and every triangular generator has value  $-2$ . These log-ratios belong to the relative interiors in Lemma 4.1. For the five-point support, this follows from  $e^{-2} + e^{-2} < 2$ . For full support, the ratio-log triple  $(-2, -2, -2)$  satisfies all principal-minor inequalities strictly and satisfies  $e^{-6} + 2 - 3e^{-2} > 0$ . Hence it lies in  $\text{int}(\mathcal{C})$ . The zero-dimensional factor also contains its unique point in its relative interior. Since relative interiors commute with finite products,  $\rho_S(a^\circ) \in \text{ri}(C_S) \cap L_S$ .  $\square$

Define the global feasible set of local log-ratio data by

$$\mathcal{K}_S := \left\{ \rho_S((\log c_\alpha)_{\alpha \in S}) : \sum_{\alpha \in S} c_\alpha \frac{\mathbf{x}^\alpha}{\alpha!} \in \mathcal{L}_S^+ \right\} \subseteq \mathbb{R}^{I(S)}.$$

LEMMA 4.5 (The feasible set for arbitrary support). *We have  $\mathcal{K}_S = C_S \cap L_S$ .*

*Proof.* If  $f \in \mathcal{L}_S^+$  and  $a_\alpha = \log c_\alpha$ , every slice is Lorentzian. Hence  $\rho_S(a) \in C_S \cap L_S$ .

Conversely, let  $r \in C_S \cap L_S$ . Choose  $a = (a_\alpha)_{\alpha \in S}$  with  $\rho_S(a) = r$ , set  $c_\alpha = e^{a_\alpha}$ , and let  $f$  be the corresponding factorially normalized polynomial. For every  $\beta$ , the local log-ratio vector of  $H_\beta(f)$  is the  $\beta$ -block  $r_\beta \in \mathcal{C}_{S_\beta}$ . The explicit criteria in Lemma 3.3 show that every slice is Lorentzian. Since  $f$  has support  $S$ , we obtain  $f \in \mathcal{L}_S^+$  and  $r \in \mathcal{K}_S$ .  $\square$

For  $\mu \in \mathbb{R}^{I(S)}$ , set

$$w_\mu := \rho_S^*(\mu) = \sum_{(\beta, \ell) \in I(S)} \mu_{\beta, \ell} \tilde{g}_{S_{\beta, \ell}}^\beta \in \mathbb{R}^S.$$

If  $\lambda \in \mathbb{R}_{\geq 0}^{I(S)}$ , then  $w_\lambda \in \text{BR}(\mathcal{L}_S^+)$  by Theorem 3.6. Its monomial is the corresponding weighted product of the local ratios. Write  $\lambda_\beta = (\lambda_{\beta, 1}, \dots, \lambda_{\beta, m(S_\beta)})$ , and let  $H_S(\lambda)$  be the logarithm of the corresponding global sharp constant:

$$H_S(\lambda) := \log B_{\mathcal{L}_S^+}(w_\lambda) = \log \sup_{f \in \mathcal{L}_S^+} \prod_{(\beta, \ell) \in I(S)} R_{\beta, \ell}(f)^{\lambda_{\beta, \ell}}.$$

**THEOREM 4.6** (Optimal bounding constants for arbitrary support). *For every  $\lambda \in \mathbb{R}_{\geq 0}^{I(S)}$ ,*

$$H_S(\lambda) = \inf_{\eta \in L_S^\perp} \sum_{\beta \in \Delta_3^{d-2}} h_{S_\beta}(\lambda_\beta + \eta_\beta),$$

where  $\eta_\beta$  denotes the  $\beta$ -block of  $\eta \in \mathbb{R}^{I(S)}$ .

*Proof.* If  $r$  denotes the local log-ratio data of  $f$ , then  $\log R_{w_\lambda}(f) = \langle \lambda, r \rangle$ , and hence  $H_S(\lambda) = \sigma_{\mathcal{K}_S}(\lambda)$ . Lemmas 4.5 and 4.4 identify  $\mathcal{K}_S = C_S \cap L_S$  and verify the relative-interior hypothesis. Lemmas 4.1 and 3.11 show that  $C_S$  is closed and convex. Applying Lemma 4.3 and using the product structure of  $C_S$  gives

$$H_S(\lambda) = \inf_{\eta \in L_S^\perp} \sigma_{C_S}(\lambda + \eta) = \inf_{\eta \in L_S^\perp} \sum_{\beta \in \Delta_3^{d-2}} h_{S_\beta}(\lambda_\beta + \eta_\beta).$$

□

*Remark 4.7.* Since  $L_S^\perp = \ker(\rho_S^*)$ , every vector  $\lambda + \eta$  in the infimum represents the same global exponent as  $\lambda$ . Thus the formula minimizes the sum of the local sharp bounds over all local representations of the same global ratio.

For full support, write  $I_d := \Delta_3^{d-2} \times [3]$ ,  $\rho_d := \rho_{\Delta_3^d}$ ,  $L_d := \text{im}(\rho_d)$ , and  $H_d := H_{\Delta_3^d}$ . Every slice then has full support and support function  $h$ .

**COROLLARY 4.8** (Full support). *For every  $\lambda = (\lambda_{\beta,i})_{(\beta,i) \in I_d} \in \mathbb{R}_{\geq 0}^{I_d}$ ,*

$$H_d(\lambda) = \inf_{\eta \in L_d^\perp} \sum_{\beta \in \Delta_3^{d-2}} h(\lambda_{\beta,1} + \eta_{\beta,1}, \lambda_{\beta,2} + \eta_{\beta,2}, \lambda_{\beta,3} + \eta_{\beta,3}).$$

*Proof.* This is Theorem 4.6 with  $S = \Delta_3^d$ .

□

## 5. Compatibility constraints

The full-support formula in Corollary 4.8 depends on the compatibility space  $L_d^\perp$ . We first compute its dimension and then exhibit all of its relations.

### 5.1. Number of compatibility constraints.

**LEMMA 5.1** (Kernel of the ratio map). *For  $d \geq 2$ ,*

$$\ker(\rho_d) = \left\{ (c_1\alpha_1 + c_2\alpha_2 + c_3\alpha_3)_{\alpha \in \Delta_3^d} : c_1, c_2, c_3 \in \mathbb{R} \right\}.$$

*In particular,  $\dim \ker(\rho_d) = 3$ .*

*Proof.* Every displayed function lies in  $\ker(\rho_d)$  because rhombus second differences annihilate linear functions.

Conversely, suppose that  $\rho_d(a) = 0$ . Write

$$b_{i,j} := a_{(i,j,d-i-j)} \quad (i, j \geq 0, i+j \leq d),$$

and, for  $i+j \leq d-1$ , set

$$P_{i,j} := b_{i+1,j} - b_{i,j}, \quad Q_{i,j} := b_{i,j+1} - b_{i,j}.$$

For  $i+j \leq d-2$ , the three equations at  $\beta = (i, j, d-2-i-j)$  give

$$P_{i,j+1} = P_{i,j}, \quad Q_{i+1,j} = Q_{i,j}, \quad P_{i+1,j} = P_{i,j+1}, \quad Q_{i,j+1} = Q_{i+1,j}.$$

The equations make all  $P_{i,j}$  equal to a common value  $p$  and all  $Q_{i,j}$  equal to a common value  $q$ . Hence  $b_{i,j} = b_{0,0} + pi + qj$ . Since  $\alpha_1 + \alpha_2 + \alpha_3 = d$ , setting  $c_3 = b_{0,0}/d$ ,  $c_1 = p + c_3$ , and  $c_2 = q + c_3$  gives  $a_\alpha = c_1\alpha_1 + c_2\alpha_2 + c_3\alpha_3$ .  $\square$

COROLLARY 5.2 (Number of independent compatibility constraints). *For  $d \geq 2$ ,*

$$\dim L_d^\perp = (d-1)(d-2).$$

*Proof.* Lemma 5.1 and rank-nullity give  $\dim L_d = \binom{d+2}{2} - 3$ , whereas  $\dim \mathbb{R}^{I_d} = 3|\Delta_3^{d-2}| = 3\binom{d}{2}$ . Hence

$$\dim L_d^\perp = 3\binom{d}{2} - \left( \binom{d+2}{2} - 3 \right) = (d-1)(d-2).$$

$\square$

5.2. *Explicit compatibility relations.* For  $d \geq 3$ , set  $J_d := \Delta_3^{d-3} \times \{1, 2\}$  and define  $\kappa_d : \mathbb{R}^{I_d} \rightarrow \mathbb{R}^{J_d}$  by

$$\begin{aligned} (\kappa_d r)_{\gamma,1} &= r_{\gamma+e_3,1} - r_{\gamma+e_2,1} + r_{\gamma+e_1,3} - r_{\gamma+e_2,3}, \\ (\kappa_d r)_{\gamma,2} &= r_{\gamma+e_3,2} - r_{\gamma+e_1,2} + r_{\gamma+e_2,3} - r_{\gamma+e_1,3}, \end{aligned}$$

for  $\gamma \in \Delta_3^{d-3}$ .

THEOREM 5.3 (Explicit compatibility theorem). *For  $d \geq 3$ ,*

$$L_d = \ker(\kappa_d).$$

*Equivalently,  $r = (r_{\beta,i})_{(\beta,i) \in I_d} \in \mathbb{R}^{I_d}$  arises from a global coefficient-log array exactly when  $\kappa_d r = 0$ . The  $2|\Delta_3^{d-3}| = (d-1)(d-2)$  displayed relations are linearly independent. Moreover, there is an exact sequence*

$$0 \longrightarrow \mathbb{R}^3 \xrightarrow{\iota_d} \mathbb{R}^{\Delta_3^d} \xrightarrow{\rho_d} \mathbb{R}^{I_d} \xrightarrow{\kappa_d} \mathbb{R}^{J_d} \longrightarrow 0,$$

where

$$\iota_d(c_1, c_2, c_3) = (c_1\alpha_1 + c_2\alpha_2 + c_3\alpha_3)_{\alpha \in \Delta_3^d}.$$

*Proof.* Let  $V_d = \mathbb{R}^{\Delta_3^d}$ , and let  $g_{\beta,i} \in V_d$  be the exponent vector of  $R_{\beta,i}$ . Define

$$\Psi_d : \mathbb{R}^{I_d} \longrightarrow V_d, \quad \Psi_d(\mathbf{e}_{\beta,i}) = g_{\beta,i}.$$

Because  $(\rho_d(a))_{\beta,i} = \langle g_{\beta,i}, a \rangle$ , we have

$$\rho_d = \Psi_d^*, \quad L_d^\perp = \ker(\Psi_d).$$

Identify  $V_d$  with the homogeneous degree- $d$  polynomials via  $\mathbf{e}_\alpha \mapsto \mathbf{x}^\alpha$ . Then  $g_{\beta,i} \leftrightarrow \mathbf{x}^\beta Q_i$ , where

$$\begin{aligned} Q_1 &= (x_1 - x_2)(x_1 - x_3), & Q_2 &= (x_2 - x_1)(x_2 - x_3), \\ Q_3 &= (x_3 - x_1)(x_3 - x_2). \end{aligned}$$

With  $u = x_1 - x_2$  and  $v = x_2 - x_3$ , these become

$$Q_1 = u(u + v), \quad Q_2 = -uv, \quad Q_3 = v(u + v).$$

A relation among the  $g_{\beta,i}$  has the form

$$F_1 Q_1 + F_2 Q_2 + F_3 Q_3 = 0,$$

where  $F_1, F_2, F_3$  are homogeneous of degree  $d - 2$ . Reducing this relation modulo  $u$  and modulo  $v$  shows, respectively, that  $u \mid F_3$  and  $v \mid F_1$ . Thus  $F_1 = vA$  and  $F_3 = uC$  for homogeneous polynomials  $A, C$  of degree  $d - 3$ , and substitution gives  $F_2 = (A + C)(u + v)$ . Hence

$$(F_1, F_2, F_3) = -A(-v, 0, u) - (A + C)(0, -(u + v), -u),$$

so every relation is generated by  $s_1 = (-v, 0, u)$  and  $s_2 = (0, -(u + v), -u)$ . These two syzygies are independent over  $\mathbb{R}[x_1, x_2, x_3]$ : the first two components of  $Ms_1 + Ns_2 = 0$  force  $M = N = 0$ . Consequently,

$$\{\mathbf{x}^\gamma s_1, \mathbf{x}^\gamma s_2 : \gamma \in \Delta_3^{d-3}\}$$

is a basis of  $\ker(\Psi_d)$ .

In the original indexing, the two relations are

$$\begin{aligned} g_{\gamma+e_3,1} - g_{\gamma+e_2,1} + g_{\gamma+e_1,3} - g_{\gamma+e_2,3} &= 0, \\ g_{\gamma+e_3,2} - g_{\gamma+e_1,2} + g_{\gamma+e_2,3} - g_{\gamma+e_1,3} &= 0. \end{aligned}$$

These are the rows of  $\kappa_d$ , so  $\text{row}(\kappa_d) = L_d^\perp$ . Their independence gives  $\ker(\kappa_d) = L_d$ . Since their number is  $2|\Delta_3^{d-3}| = \dim \mathbb{R}^{J_d}$ , the map  $\kappa_d$  is surjective. Finally, Lemma 5.1 identifies  $\ker(\rho_d) = \text{im}(\iota_d)$ , and  $\iota_d$  is injective. This proves exactness.  $\square$

Theorem 5.3 replaces the abstract constraint  $\eta \in L_d^\perp$  with explicit variables. We adopt the boundary convention  $\mu_\alpha = \nu_\alpha = 0$  for  $\alpha \notin \Delta_3^{d-3}$ . For  $\mu, \nu \in \mathbb{R}^{\Delta_3^{d-3}}$ , the adjoint  $\kappa_d^* : \mathbb{R}^{J_d} \rightarrow \mathbb{R}^{I_d}$  is

$$\begin{aligned} (\kappa_d^*(\mu, \nu))_{\beta,1} &= \mu_{\beta-e_3} - \mu_{\beta-e_2}, \\ (\kappa_d^*(\mu, \nu))_{\beta,2} &= \nu_{\beta-e_3} - \nu_{\beta-e_1}, \\ (\kappa_d^*(\mu, \nu))_{\beta,3} &= \mu_{\beta-e_1} - \mu_{\beta-e_2} + \nu_{\beta-e_2} - \nu_{\beta-e_1}. \end{aligned}$$

COROLLARY 5.4 (Explicit arbitrary-degree formula). *Let  $d \geq 3$  and  $\lambda = (\lambda_{\beta,i})_{(\beta,i) \in I_d} \in \mathbb{R}_{\geq 0}^{I_d}$ . Then*

$$\begin{aligned} H_d(\lambda) = \inf_{\mu, \nu \in \mathbb{R}^{\Delta_3^{d-3}}} \sum_{\beta \in \Delta_3^{d-2}} h \Big( & \lambda_{\beta,1} + \mu_{\beta-e_3} - \mu_{\beta-e_2}, \\ & \lambda_{\beta,2} + \nu_{\beta-e_3} - \nu_{\beta-e_1}, \\ & \lambda_{\beta,3} + \mu_{\beta-e_1} - \mu_{\beta-e_2} + \nu_{\beta-e_2} - \nu_{\beta-e_1} \Big). \end{aligned}$$

*Proof.* Theorem 5.3 gives  $L_d^\perp = \text{im}(\kappa_d^*)$ . Substituting  $\eta = \kappa_d^*(\mu, \nu)$  into Corollary 4.8 gives the formula.  $\square$

## 6. The full-support cubic case

A full-support cubic has three quadratic slices and nine local ratios. Corollary 5.2 gives  $\dim L_3^\perp = 2$ , allowing an explicit description of both compatibility and the sharp constants.

6.1. *Cubic compatibility and optimal bounding constants.* For

$$f(\mathbf{x}) = \sum_{\alpha \in \Delta_3^3} c_\alpha \frac{\mathbf{x}^\alpha}{\alpha!} \in \mathcal{L}_{3,3}^+,$$

write  $c_{ijk} := c_{(i,j,k)}$  and abbreviate  $R_{e_i,j}$  by  $R_{ij}$ . The nine ratios are

$$\begin{aligned} R_{11} &= \frac{c_{111}c_{300}}{c_{210}c_{201}}, & R_{12} &= \frac{c_{201}c_{120}}{c_{210}c_{111}}, & R_{13} &= \frac{c_{210}c_{102}}{c_{201}c_{111}}, \\ R_{21} &= \frac{c_{021}c_{210}}{c_{120}c_{111}}, & R_{22} &= \frac{c_{111}c_{030}}{c_{120}c_{021}}, & R_{23} &= \frac{c_{120}c_{012}}{c_{111}c_{021}}, \\ R_{31} &= \frac{c_{012}c_{201}}{c_{111}c_{102}}, & R_{32} &= \frac{c_{102}c_{021}}{c_{111}c_{012}}, & R_{33} &= \frac{c_{111}c_{003}}{c_{102}c_{012}}. \end{aligned}$$

The  $i$ th row  $(R_{i1}, R_{i2}, R_{i3})$  records the ratios of  $H_{e_i}(f)$ .

Let

$$\mathcal{R}_{\text{loc}} := \{(X, Y, Z) \in \mathbb{R}_{>0}^3 : XY, XZ, YZ \leq 1, \quad XYZ + 2 - X - Y - Z \geq 0\}.$$

This set is the full-support quadratic feasible region in ratio coordinates, and  $\mathcal{C}$  is its coordinatewise logarithm.

LEMMA 6.1 (Cubic compatibility and lifting). *Let  $R_{ij} > 0$  for  $1 \leq i, j \leq 3$ .*

- (i) *The nine numbers  $R_{ij}$  arise from a positive cubic coefficient array if and only if*

$$R_{13}R_{31} = R_{21}R_{23}, \quad R_{23}R_{32} = R_{12}R_{13}.$$

- (ii) *They arise from a polynomial  $f \in \mathcal{L}_{3,3}^+$  if and only if the two relations in (i) hold and  $(R_{i1}, R_{i2}, R_{i3}) \in \mathcal{R}_{\text{loc}}$  for  $i = 1, 2, 3$ .*

*Proof.* Set  $r_{ij} = \log R_{ij}$ . Theorem 5.3 gives  $r \in L_3 = \text{im}(\rho_3)$  exactly when

$$r_{13} + r_{31} - r_{21} - r_{23} = 0, \quad r_{23} + r_{32} - r_{12} - r_{13} = 0.$$

Exponentiating yields the relations in (i), while membership in  $\text{im}(\rho_3)$  is precisely the existence of a coefficient-log array with these ratios.

For (ii), Lemma 3.3 makes a slice Lorentzian exactly when its row belongs to  $\mathcal{R}_{\text{loc}}$ . The Hessian-slice characterization then proves the claim.  $\square$

The two compatibility relations also give  $R_{12}R_{21} = R_{31}R_{32}$ .

We order the ratios by

$$(R_1, \dots, R_9) := (R_{11}, R_{12}, R_{13}, R_{21}, R_{22}, R_{23}, R_{31}, R_{32}, R_{33}),$$

and use the same indexing for  $\lambda = (\lambda_1, \dots, \lambda_9)$ .

**THEOREM 6.2** (Cubic optimal bounding-constant formula). *For every  $\lambda \in \mathbb{R}_{\geq 0}^9$ ,*

$$\begin{aligned} H_3(\lambda) = \inf_{\mu, \nu \in \mathbb{R}} & \left[ h(\lambda_1, \lambda_2 - \nu, \lambda_3 + \mu - \nu) \right. \\ & + h(\lambda_4 - \mu, \lambda_5, \lambda_6 - \mu + \nu) \\ & \left. + h(\lambda_7 + \mu, \lambda_8 + \nu, \lambda_9) \right]. \end{aligned}$$

*Proof.* For  $d = 3$ , the two variables in Corollary 5.4 reduce to scalars  $\mu$  and  $\nu$ . They change the three rows by

$$(0, -\nu, \mu - \nu), \quad (-\mu, 0, -\mu + \nu), \quad (\mu, \nu, 0).$$

Substituting these corrections into Corollary 5.4 proves the formula.  $\square$

**6.2. Two-generator sections.** We now determine the sharp constant on every two-generator section of the cubic bounded-ratio cone.

**THEOREM 6.3** (Two-generator classification). *Let  $R_{ij}$  and  $R_{k\ell}$  be two distinct cubic local ratios, and let  $a, b \geq 0$ . Then exactly one of the following cases occurs.*

- (i) *If the two ratios lie in the same row, then*

$$\sup_{f \in \mathcal{L}_{3,3}^+} R_{ij}(f)^a R_{k\ell}(f)^b = 2^{a+b} \frac{a^a b^b}{(a+b)^{a+b}}.$$

(ii) If they form a transposed off-diagonal pair  $R_{ij}, R_{ji}$  with  $i \neq j$ , then

$$\sup_{f \in \mathcal{L}_{3,3}^+} R_{ij}(f)^a R_{ji}(f)^b = 2^{|a-b|}.$$

(iii) In every other case,

$$\sup_{f \in \mathcal{L}_{3,3}^+} R_{ij}(f)^a R_{kl}(f)^b = 2^{a+b}.$$

Here  $0^0 = 1$ .

*Proof.* We first show that any locally feasible row extends to a Lorentzian cubic. Fix  $(X, Y, Z) \in \mathcal{R}_{\text{loc}}$ , choose  $\varepsilon > 0$  with  $\varepsilon \leq 1$  and  $\varepsilon Z \leq 1$ , and take the other two rows to be

$$(\varepsilon Z, 1, 1), \quad (\varepsilon, YZ, 1).$$

Their determinant expressions are 0 and  $(1 - \varepsilon)(1 - YZ) \geq 0$ , respectively, and all pairwise products are at most 1. The identities  $Z\varepsilon = (\varepsilon Z) \cdot 1$  and  $YZ = Y \cdot Z$  verify compatibility. Lemma 6.1 therefore realizes the prescribed row.

We will also use several explicit degenerations. For  $0 < t \leq 1$ , set  $A_t := 2/(1 + t)$  and define

$$\begin{aligned} E_1(t) &= (A_t, t, t), & E_2(t) &= (t, A_t, t), & E_3(t) &= (t, t, A_t), \\ F(t) &= (A_t^{-1}, t, A_t t), & G(t) &= (1, 1, t), & K(t) &= \left(A_t, \frac{t^2}{A_t}, t\right). \end{aligned}$$

All six rows belong to  $\mathcal{R}_{\text{loc}}$ . The determinant expression vanishes on  $E_i(t)$  and  $G(t)$ , while its values on  $F(t)$  and  $K(t)$  are

$$\frac{(1 - t)^2(2t + 3)}{2(1 + t)} \quad \text{and} \quad \frac{t(1 - t)^2(t + 2)}{2(1 + t)},$$

respectively. Both values are nonnegative, and the product inequalities follow by substitution.

For (i), denote the selected ratios by  $X, Y$  and the remaining ratio by  $Z$ . Local feasibility gives

$$0 \leq XYZ + 2 - X - Y - Z = 2 - X - Y - Z(1 - XY),$$

so  $XY \leq 1$  implies  $X + Y \leq 2$ .

Conversely, suppose that  $X, Y > 0$  and  $X + Y < 2$ . Then  $XY < 1$ , and any choice of

$$0 < Z \leq \min \left\{ X^{-1}, Y^{-1}, \frac{2 - X - Y}{1 - XY} \right\}$$

produces a row in  $\mathcal{R}_{\text{loc}}$ . The extension above realizes this row in a Lorentzian cubic. Continuity therefore reduces the desired supremum to

$$\sup_{\substack{X, Y > 0 \\ X+Y \leq 2}} X^a Y^b.$$

For  $a, b > 0$ , the maximum occurs at  $(X, Y) = (2a/(a+b), 2b/(a+b))$  and equals the value in (i). Continuity covers zero exponents.

For (ii), it suffices by symmetry to consider  $(R_{12}, R_{21})$ . Compatibility gives  $R_{12}R_{21} = R_{31}R_{32} \leq 1$ , while each factor is at most 2. If  $a \geq b$ , then  $R_{12}^a R_{21}^b = R_{12}^{a-b} (R_{12}R_{21})^b \leq 2^{a-b}$ . The case  $b \geq a$  is symmetric.

Suppose  $a \geq b$ . The rows  $E_2(t), F(t), G(t)$  are locally feasible and satisfy the two compatibility relations with common values  $t$  and  $A_t t$ . Lemma 6.1 realizes them. As  $t \rightarrow 0^+$ , the selected ratios tend to 2 and  $1/2$ , so their weighted product tends to  $2^{a-b}$ . Interchanging the indices proves sharpness when  $b \geq a$ .

For (iii), the individual bounds give  $R_{ij}^a R_{kl}^b \leq 2^{a+b}$ . It remains to approach equality. Up to a simultaneous permutation of the indices, the admissible pairs have five representatives:

$$\{R_{11}, R_{21}\}, \quad \{R_{11}, R_{22}\}, \quad \{R_{12}, R_{23}\}, \quad \{R_{12}, R_{32}\}, \quad \{R_{11}, R_{23}\}.$$

Indeed, two diagonal ratios form one type. After excluding same-row pairs, a pair with exactly one diagonal ratio is represented by either  $\{R_{11}, R_{21}\}$  or  $\{R_{11}, R_{23}\}$ . After also excluding transposed pairs, two off-diagonal ratios either share a target or form a directed path.

For these representatives, choose the three rows in the following table. Here  $C_1$  and  $C_2$  denote the common values in the first and second compatibility relations.

| selected pair        | row 1    | row 2    | row 3    | $(C_1, C_2)$   |
|----------------------|----------|----------|----------|----------------|
| $\{R_{11}, R_{21}\}$ | $E_1(t)$ | $E_1(t)$ | $E_1(t)$ | $(tA_t, t^2)$  |
| $\{R_{11}, R_{22}\}$ | $E_1(t)$ | $E_2(t)$ | $E_3(t)$ | $(t^2, t^2)$   |
| $\{R_{12}, R_{23}\}$ | $E_2(t)$ | $E_3(t)$ | $E_1(t)$ | $(tA_t, tA_t)$ |
| $\{R_{12}, R_{32}\}$ | $E_2(t)$ | $E_2(t)$ | $E_2(t)$ | $(t^2, tA_t)$  |
| $\{R_{11}, R_{23}\}$ | $E_1(t)$ | $E_3(t)$ | $K(t)$   | $(tA_t, t^2)$  |

Each line of the table gives three locally feasible and compatible rows, so Lemma 6.1 realizes them. Both selected ratios equal  $A_t$  and tend to 2 as  $t \rightarrow 0^+$ . This proves sharpness in the remaining cases.  $\square$

## 7. Applications and extremal comparisons

We now compare the universal Lorentzian bounds with two structured subclasses. Volume polynomials attain one Lorentzian optimum but exhibit a



strict gap for another functional. Rank-three matroids satisfy an even stronger inequality through the Rayleigh property.

7.1. *A Lorentzian extremum realized by mixed volumes.* For a full-support Lorentzian cubic, define

$$\Phi_{\text{diag}}(f) := R_{11}R_{22}R_{33} = \frac{c_{111}^3 c_{300} c_{030} c_{003}}{c_{210} c_{201} c_{120} c_{021} c_{102} c_{012}}.$$

Let  $\mathcal{V}_3^3$  be the full-support volume polynomials

$$F_K(x_1, x_2, x_3) = \text{Vol}(x_1 K_1 + x_2 K_2 + x_3 K_3)$$

of triples of convex bodies in  $\mathbb{R}^3$ . They are Lorentzian by [4, Theorem 4.1]. In factorial normalization, Minkowski's polynomial formula identifies  $c_\alpha$  with six times the mixed volume containing  $\alpha_i$  copies of  $K_i$  for each  $i \in [3]$  [9, Section 5.1].

PROPOSITION 7.1 (No geometric loss). *We have*

$$\sup_{F \in \mathcal{V}_3^3} \Phi_{\text{diag}}(F) = \sup_{f \in \mathcal{L}_{3,3}^+} \Phi_{\text{diag}}(f) = 8.$$

*Proof.* The local bound  $R_{ii} \leq 2$  gives  $\Phi_{\text{diag}} \leq 8$ .

To prove sharpness among volume polynomials, let  $e_1, e_2, e_3$  be the standard basis of  $\mathbb{R}^3$ . For  $0 < \varepsilon \leq 1/2$ , set

$$P_1^{(\varepsilon)} = \text{conv}\{0, e_1, \varepsilon e_2, \varepsilon e_3, \varepsilon(e_2 + e_3)\},$$

and obtain  $P_2^{(\varepsilon)}$  and  $P_3^{(\varepsilon)}$  by cyclically permuting the coordinates. We write  $P_i = P_i^{(\varepsilon)}$  below.

These square pyramids satisfy

$$\text{Vol}(P_i) = \frac{\varepsilon^2}{3}, \quad V(P_i, P_i, P_j) = \frac{\varepsilon + \varepsilon^2 + \varepsilon^3}{6} \quad (i \neq j),$$

and

$$V(P_1, P_2, P_3) = \frac{1}{6}.$$

The first identity follows from the pyramid-volume formula. By symmetry, it remains to compute  $V(P_1, P_1, P_2)$ . In the polytope mixed-area formula, only the two noncoordinate triangular facets of  $P_1$  contribute [9, Section 5.1]. Their outward unit normals are  $n_2 = (\varepsilon, 1, 0)/\sqrt{1 + \varepsilon^2}$  and  $n_3 = (\varepsilon, 0, 1)/\sqrt{1 + \varepsilon^2}$ , and each facet has area  $\frac{\varepsilon}{2}\sqrt{1 + \varepsilon^2}$ . The corresponding support values of  $P_2$  are  $1/\sqrt{1 + \varepsilon^2}$  and  $(\varepsilon + \varepsilon^2)/\sqrt{1 + \varepsilon^2}$ . The coordinate facets contribute zero, so the formula gives

$$3V(P_1, P_1, P_2) = \frac{\varepsilon}{2} + \frac{\varepsilon(\varepsilon + \varepsilon^2)}{2}.$$

To obtain the final mixed volume, put  $S_i = [0, e_i]$  and  $\Delta = \text{conv}\{0, e_1, e_2, e_3\}$ . Since  $S_i \subseteq P_i \subseteq \Delta$  and  $V(S_1, S_2, S_3) = V(\Delta, \Delta, \Delta) = 1/6$ , monotonicity forces  $V(P_1, P_2, P_3) = 1/6$ .

These identities yield

$$\Phi_{\text{diag}}(F_P) = \frac{8}{(1 + \varepsilon + \varepsilon^2)^6} \longrightarrow 8 \quad (\varepsilon \rightarrow 0^+).$$

Thus volume polynomials asymptotically attain the Lorentzian optimum.  $\square$

7.2. *A strict mixed-volume gap.* Consider the coefficient functional

$$\Psi(f) := \frac{c_{201}c_{102}c_{030}}{c_{111}^3} = R_{12}R_{21}R_{22}R_{23}R_{32}.$$

PROPOSITION 7.2 (A strict geometric gap). *Set*

$$C_* := \frac{32}{27}(13\sqrt{13} - 46) = 1.0336789108\dots$$

Then

$$\sup_{f \in \mathcal{L}_{3,3}^+} \Psi(f) = C_*, \quad \sup_{F \in \mathcal{V}_3^3} \Psi(F) = 1.$$

In particular, the mixed-volume optimum is strictly smaller than the Lorentzian optimum.

*Proof.* We first compute the Lorentzian optimum. Put  $x = R_{13}$ ,  $y = R_{31}$ ,  $p = R_{21}$ , and  $q = R_{23}$ . By cubic compatibility,

$$xy = pq =: t^2, \quad 0 < t \leq 1.$$

The determinant condition for a feasible row gives  $2 - X - Y \geq Z(1 - XY) \geq 0$ . Applying it to the first and third rows yields  $R_{12} \leq 2 - x$  and  $R_{32} \leq 2 - y$ . Hence, by AM–GM,

$$R_{12}R_{32} \leq (2 - x)(2 - y) \leq (2 - t)^2.$$

The middle row  $(p, R_{22}, q)$  gives, for  $t < 1$ ,

$$R_{22} \leq \frac{2 - p - q}{1 - t^2} \leq \frac{2}{1 + t}.$$

When  $t = 1$ , feasibility forces  $p = q = 1$  and again gives  $R_{22} \leq 1$ . Therefore

$$\Psi(f) \leq \phi(t) := \frac{2t^2(2 - t)^2}{1 + t}, \quad 0 < t \leq 1.$$

Its derivative is

$$\phi'(t) = \frac{2t(t - 2)(3t^2 + 2t - 4)}{(1 + t)^2},$$

which has the unique maximizer  $t_* = (\sqrt{13} - 1)/3$  on  $(0, 1]$ . At this point,  $\phi(t_*) = C_*$ .

To approach this bound, fix  $t = t_*$  and  $A = 2/(1+t)$ . For sufficiently small  $\varepsilon > 0$ , put  $b_\varepsilon = 2 - t - \varepsilon$  and  $s_\varepsilon = \varepsilon/2$ , and take the three ratio rows

$$(s_\varepsilon, b_\varepsilon, t), \quad (t, A, t), \quad (t, b_\varepsilon, s_\varepsilon).$$

For small  $\varepsilon$ , the first and third rows are strictly feasible and the middle row has determinant zero. The compatibility identities are

$$R_{13}R_{31} = t^2 = R_{21}R_{23}, \quad R_{23}R_{32} = tb_\varepsilon = R_{12}R_{13}.$$

Lemma 6.1 realizes these data by  $f_\varepsilon \in \mathcal{L}_{3,3}^+$ , and

$$\Psi(f_\varepsilon) = \frac{2t^2b_\varepsilon^2}{1+t} \longrightarrow C_*.$$

It remains to optimize over volume polynomials. Let  $A, B, C \subseteq \mathbb{R}^3$  be convex bodies. The inequality  $\Psi \leq 1$  is equivalent to

$$V(A, A, C)V(A, C, C)\text{Vol}(B) \leq V(A, B, C)^3.$$

Approximation and continuity reduce the proof to bodies with nonempty interior.

The mixed-body theorem provides a convex body  $K = [A, C]$ , unique up to translation, for which

$$V(L, K, K) = V(L, A, C)$$

for every convex body  $L \subseteq \mathbb{R}^3$  [6] and [9, Sections 5.1 and 7.3]. Taking  $L = K$  and  $L = C$  gives

$$V(A, K, C) = \text{Vol}(K), \quad V(K, K, C) = V(A, C, C).$$

Applying the Alexandrov–Fenchel inequality to  $A, K, C$  yields

$$\text{Vol}(K)^2 = V(A, K, C)^2 \geq V(A, A, C)V(K, K, C) = V(A, A, C)V(A, C, C).$$

The choice  $L = B$  gives

$$V(A, B, C) = V(B, K, K).$$

Minkowski's first inequality then implies

$$V(A, B, C)^3 = V(B, K, K)^3 \geq \text{Vol}(B)\text{Vol}(K)^2.$$

Combining these estimates proves the required mixed-volume inequality. Thus every volume polynomial satisfies  $\Psi \leq 1$ . Taking  $A = B = C$  gives equality, so the constant 1 is sharp.  $\square$

Consequently, a full-support Lorentzian cubic with  $c_{201}c_{102}c_{030} > c_{111}^3$  cannot be a volume polynomial.

7.3. *A strict combinatorial improvement.* Let  $M$  be a rank-three matroid on

$$E = E_1 \sqcup E_2 \sqcup E_3,$$

and set

$$b_{abc} = \#\{B \in \mathcal{B}(M) : |B \cap E_1| = a, |B \cap E_2| = b, |B \cap E_3| = c\}.$$

The grouped basis polynomial

$$B_{M,E}(x_1, x_2, x_3) = \sum_{a+b+c=3} b_{abc} x_1^a x_2^b x_3^c$$

is Lorentzian: the basis-generating polynomial is Lorentzian by [4, Theorem 3.10], and identifying variables within each color class preserves Lorentzianity by [4, Theorem 2.10].

In factorial normalization,

$$c_{abc} = a!b!c!b_{abc}.$$

Hence the Lorentzian bound  $\Phi_{\text{diag}} \leq 8$ , extended to arbitrary support by density [4, Theorem 2.25], gives

$$b_{111}^3 b_{300} b_{030} b_{003} \leq \frac{64}{27} b_{210} b_{201} b_{120} b_{021} b_{102} b_{012}.$$

For rank-three matroids, the Rayleigh property improves the constant  $\frac{64}{27}$  to 1.

**PROPOSITION 7.3** (Rayleigh improvement). *For every rank-three matroid and every three-coloring,*

$$b_{111}^3 b_{300} b_{030} b_{003} \leq b_{210} b_{201} b_{120} b_{021} b_{102} b_{012}.$$

*The constant 1 is sharp.*

*Proof.* By [10, Theorem 1.1], every rank-three matroid is Rayleigh. Write

$$Z_M(\mathbf{y}) = \sum_{B \in \mathcal{B}(M)} \prod_{e \in B} y_e, \quad \Delta_{ef} Z_M = (\partial_e Z_M)(\partial_f Z_M) - Z_M \partial_{ef} Z_M.$$

Thus  $\Delta_{ef} Z_M \geq 0$  on the positive orthant. After specializing  $y_e = x_i$  for  $e \in E_i$ ,

$$(\partial_2 B_{M,E})(\partial_3 B_{M,E}) - B_{M,E} \partial_{23} B_{M,E} = \sum_{e \in E_2} \sum_{f \in E_3} \Delta_{ef} Z_M \Big|_{y_g = x_i, g \in E_i} \geq 0.$$

Setting  $x_2 = x_3 = 0$  by continuity gives

$$x_1^4 (b_{210} b_{201} - b_{300} b_{111}) \geq 0,$$

hence

$$b_{300} b_{111} \leq b_{210} b_{201}.$$

Multiplying this inequality with its two cyclic analogues proves the claim.

For sharpness, take

$$M_N = U_{1,N+2} \oplus U_{1,N+2} \oplus U_{1,N+2},$$

with color multiplicities

$$(N, 1, 1), \quad (1, N, 1), \quad (1, 1, N)$$

in the three components. Then

$$B_{M_N, E}(x, y, z) = (Nx + y + z)(x + Ny + z)(x + y + Nz),$$

and therefore

$$\frac{b_{111}^3 b_{300} b_{030} b_{003}}{b_{210} b_{201} b_{120} b_{021} b_{102} b_{012}} = \frac{N^3(N^3 + 3N + 2)^3}{(N^2 + N + 1)^6} \rightarrow 1.$$

□

*Remark 7.4* (Combinatorial interpretation). Proposition 7.3 has two elementary specializations. If  $M$  is the direct sum of three rank-one uniform matroids, viewed as three boxes with three-colored elements, let  $N_{abc}$  denote the number of choices of one element from each box with color profile  $(a, b, c)$ ; then  $b_{abc} = N_{abc}$ . If  $M = M(G)$  is the graphic matroid of a connected four-vertex graph with three-colored edges, let  $t_{abc}$  denote the number of spanning trees with color profile  $(a, b, c)$ ; then  $b_{abc} = t_{abc}$ . Thus the three-box and three-color spanning-tree inequalities are two specializations of the same rank-three Rayleigh inequality.

## Appendix A. The local-to-global principle in four variables

We now prove the two four-variable results stated in Section 3. We retain the notation  $S_\beta$ ,  $\tilde{v}^\beta$ , and  $\mathcal{G}_4(S)$  introduced there. For  $\alpha, \beta \in \mathbb{N}^4$ , write  $\alpha \wedge \beta$  for their coordinatewise minimum.

**A.1. Quaternary cubics.** Theorem 3.6 proves the local-to-global theorem in three variables for every degree. We show that the same statement remains true for cubics in four variables. The only new local phenomenon is four-point branching in a quadratic slice; in degree three it can still be resolved without enumerating supports.

For convenience, we restate Theorem 3.9 before giving its proof.

**THEOREM 3.9** (Quaternary cubics). *For every nonempty  $M$ -convex  $S \subseteq \Delta_4^3$ ,*

$$\text{BR}(\mathcal{L}_S^+) = \text{Cone}(\mathcal{G}_4(S)).$$

*Equivalently, every bounded monomial ratio among the normalized coefficients of a quaternary Lorentzian cubic is generated by bounded ratios arising from its quadratic Hessian slices.*

We use one standard notion from discrete convex analysis. A function  $q : S \rightarrow \mathbb{R}$  is  $M$ -concave if, whenever  $\alpha, \beta \in S$  and  $\alpha_i > \beta_i$ , there is  $j$  with  $\alpha_j < \beta_j$  such that

$$q(\alpha) + q(\beta) \leq q(\alpha - e_i + e_j) + q(\beta + e_i - e_j).$$

On an  $M$ -convex domain it is enough to check the local exchanges  $\|\alpha - \beta\|_1 = 4$ ; see [7, Section 6.2]. We also use [4, Theorem 3.14]: if  $q$  is  $M$ -concave, then

$$(A.1) \quad f_t(\mathbf{x}) := \sum_{\alpha \in S} e^{tq(\alpha)} \frac{\mathbf{x}^\alpha}{\alpha!}$$

is Lorentzian for every  $t \geq 0$ .

### Local directions and support reduction.

We first isolate the quadratic directions used in the proof. The assertions of the following lemma also appear, in an equivalent formulation, in Baker–Huh–Kummer–Lorscheid [1, Lemma 5.15]. We include the short proof for completeness.

**LEMMA A.1** (Quadratic inequalities). *Let  $P = (p_{ij})_{1 \leq i, j \leq 4}$  be symmetric and entrywise nonnegative, with at most one positive eigenvalue. For distinct  $i, j$ ,  $p_{ii}p_{jj} \leq p_{ij}^2$ , and for pairwise distinct  $i, j, k$ ,  $p_{ii}p_{jk} \leq 2p_{ij}p_{ik}$ . If  $i, j, k, \ell$  are pairwise distinct, then*

$$\sqrt{p_{ij}p_{k\ell}}, \quad \sqrt{p_{ik}p_{j\ell}}, \quad \sqrt{p_{i\ell}p_{jk}}$$

*are the side lengths of a possibly degenerate triangle. In particular, if one of the three opposite-edge products vanishes, then the other two are equal.*

*Proof.* The principal  $2 \times 2$  submatrix on  $\{i, j\}$  gives  $p_{ii}p_{jj} \leq p_{ij}^2$ . For the second inequality, decrease the  $j$ - and  $k$ -diagonal entries of the principal submatrix on  $\{i, j, k\}$  to zero. The resulting matrix still has at most one positive eigenvalue. If  $p_{jk} > 0$ , its determinant is nonnegative, while

$$\det \begin{pmatrix} p_{ii} & p_{ij} & p_{ik} \\ p_{ij} & 0 & p_{jk} \\ p_{ik} & p_{jk} & 0 \end{pmatrix} = p_{jk}(2p_{ij}p_{ik} - p_{ii}p_{jk}).$$

The case  $p_{jk} = 0$  is immediate. The four-point assertion is [5, Proposition 3.1], extended to the nonnegative boundary by continuity. If one side length vanishes, the triangle inequalities force the other two to coincide.  $\square$

For a nonempty  $M$ -convex support  $T \subseteq \Delta_4^2$ , let  $\mathcal{E}(T) \subseteq \mathbb{R}^T$  consist of the support-defined vectors

$$(A.2) \quad \varepsilon_{2e_i} + \varepsilon_{2e_j} - 2\varepsilon_{e_i+e_j},$$

$$(A.3) \quad \varepsilon_{2e_i} + \varepsilon_{e_j+e_k} - \varepsilon_{e_i+e_j} - \varepsilon_{e_i+e_k},$$

where the indices in (A.3) are pairwise distinct, together with both signs of

$$(A.4) \quad \varepsilon_{e_i+e_j} + \varepsilon_{e_k+e_\ell} - \varepsilon_{e_i+e_k} - \varepsilon_{e_j+e_\ell}$$

whenever  $i, j, k, \ell$  are pairwise distinct, the four displayed support points are present, and at least one of  $e_i + e_\ell, e_j + e_k$  is absent. Lemma A.1 shows that every vector in  $\mathcal{E}(T)$  is a bounded-ratio direction on  $\mathcal{L}_T^+$ : the bounds in (A.2) and (A.3) are 1 and 2, while (A.4) is an equality direction.

From now on let  $S \subseteq \Delta_4^3$ . Write  $S_i := S_{e_i}$  and define

$$\mathcal{G}_4^{\text{el}}(S) := \{\tilde{g}^{e_i} : i \in [4], S_i \neq \emptyset, g \in \mathcal{E}(S_i)\}.$$

Every  $h \in \mathcal{G}_4^{\text{el}}(S)$  is balanced:

$$(A.5) \quad \sum_{\alpha \in S} h_\alpha \alpha = 0.$$

Moreover,

$$(A.6) \quad \text{Cone}(\mathcal{G}_4^{\text{el}}(S)) \subseteq \text{Cone}(\mathcal{G}_4(S)) \subseteq \text{BR}(\mathcal{L}_S^+).$$

The first inclusion follows from the definition of  $\mathcal{E}(S_i)$ ; the second follows because every quadratic Hessian slice of a Lorentzian cubic is Lorentzian.

Call a nonempty slice  $S_i$  *branching* if all six off-diagonal points  $e_j + e_k$ ,  $1 \leq j < k \leq 4$ , belong to  $S_i$ . Put  $\mathbf{1} := (1, 1, 1, 1)$  and  $z_i := \mathbf{1} - e_i$ .

LEMMA A.2 (Support reduction). *Suppose that  $S_i$  is branching. Then either  $S = e_i + T$  for some nonempty  $M$ -convex  $T \subseteq \Delta_4^2$ , or  $z_1, z_2, z_3, z_4 \in S$ .*

*Proof.* Branching of  $S_i$  gives  $z_j \in S$  for every  $j \neq i$ . If  $\alpha_i \geq 1$  for all  $\alpha \in S$ , then  $S = e_i + T$  with  $T = S - e_i$ , and  $T$  is  $M$ -convex by symmetric exchange.

Otherwise choose  $\alpha \in S$  with  $\alpha_i = 0$ . If  $\alpha = z_i$ , there is nothing to prove. Since  $|\alpha| = 3$ , there is  $j \neq i$  with  $\alpha_j \geq 2$ . Suppose first that  $\alpha = 2e_j + e_k$ , and let  $\ell$  be the remaining index. Compare  $\alpha$  with  $z_j = e_i + e_k + e_\ell$ . At the  $j$ -coordinate, the exchange axiom can use only  $i$  or  $\ell$ . In the first case,  $z_j - e_i + e_j = z_i \in S$ , while in the second case,  $\alpha - e_j + e_\ell = z_i \in S$ . If  $\alpha = 3e_j$ , compare  $\alpha$  with  $z_j$ . If the exchange uses  $i$ , then again  $z_j - e_i + e_j = z_i \in S$ . Otherwise it produces  $2e_j + e_k \in S$  for some  $k \notin \{i, j\}$ , and the preceding case applies.  $\square$

LEMMA A.3 (Translated quadratic supports). *Suppose  $S = e_i + T$  with  $T \subseteq \Delta_4^2$  nonempty and  $M$ -convex. Differentiation in  $x_i$  preserves the normalized coefficient array and*

$$f \in \mathcal{L}_S^+ \iff \partial_i f \in \mathcal{L}_T^+.$$

Consequently,  $\text{BR}(\mathcal{L}_S^+) = \text{BR}(\mathcal{L}_T^+)$  under the natural identification of coefficient coordinates.

*Proof.* In factorial normalization,

$$\partial_i \left( \sum_{\gamma \in T} c_{e_i + \gamma} \frac{\mathbf{x}^{e_i + \gamma}}{(e_i + \gamma)!} \right) = \sum_{\gamma \in T} c_{e_i + \gamma} \frac{\mathbf{x}^\gamma}{\gamma!}.$$

Thus the normalized coefficient arrays agree. If  $j \neq i$ , every nonzero quadratic derivative  $\partial_j f$  is divisible by  $x_i$ ; after placing the  $i$ th coordinate first, its Hessian has the form  $\begin{pmatrix} a & b^\top \\ b & 0 \end{pmatrix}$ . Interlacing with the zero  $3 \times 3$  principal block shows that this Hessian has at most one positive eigenvalue. The characterization [4, Theorem 2.25] gives the equivalence.  $\square$

### The branching cubic case.

Assume for the rest of this subsection that  $z_1, z_2, z_3, z_4 \in S$ . For  $i \neq j$ , set  $a_{ij} := 2e_i + e_j$ , and set  $p_i := 3e_i$ . Let  $N_i := \{j \neq i : a_{ij} \in S\}$ . Together with the  $z_i$ , the points  $a_{ij}$  and  $p_i$  exhaust  $\Delta_4^3$ . We write  $i \rightarrow j$  when  $a_{ij} \in S$  and  $i \leftrightarrow j$  when both directions occur.

LEMMA A.4 (Support graph). *For every  $i \in [4]$ ,*

$$|N_i| \in \{0, 2, 3\}, \quad p_i \in S \implies |N_i| = 3.$$

*Proof.* If  $a_{ij} \in S$  and  $k, \ell$  are the remaining indices, compare  $a_{ij}$  with  $z_j = e_i + e_k + e_\ell$ . Symmetric exchange at the  $j$ -coordinate forces  $a_{ik} \in S$  or  $a_{i\ell} \in S$ , so  $|N_i| \neq 1$ .

If  $p_i \in S$ , fix  $j \neq i$  and let  $k, \ell$  be the remaining indices. Symmetric exchange between  $p_i$  and  $z_j$  at the  $i$ -coordinate forces both  $a_{ik}$  and  $a_{i\ell}$  into  $S$ . Varying  $j$  gives  $|N_i| = 3$ .  $\square$

Let  $q = (q_\alpha)_{\alpha \in S} \in \mathbb{R}^S$ . The vectors  $z_1, \dots, z_4$  form a basis of  $\mathbb{R}^4$ , since their matrix is  $J - I$ . Hence there is a unique linear function  $\ell(\alpha) = \langle b, \alpha \rangle$  such that  $(q - \ell)(z_i) = 0$  for all  $i$ . For a normalized  $q$  with  $q(z_i) = 0$ , set  $s_{ij} := -q(a_{ij})$  when  $a_{ij} \in S$  and  $r_i := -q(p_i)$  when  $p_i \in S$ .

LEMMA A.5 (Local exchange criterion). *Assume  $q(z_i) = 0$  for all  $i$ . If  $q \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ$ , then, whenever the displayed coordinates are defined,*

$$(A.7) \quad s_{ij} \geq 0,$$

$$(A.8) \quad s_{ij} \leq s_{ik} + s_{ji},$$

$$(A.9) \quad s_{ij} + s_{ik} \leq r_i.$$

*If  $N_i = \{j, k\}$ , then*

$$(A.10) \quad s_{ij} = s_{ik}.$$



Moreover, among normalized functions satisfying (A.7)–(A.10),  $M$ -concavity is equivalent to

$$(A.11) \quad \min_{j \in N_i} s_{ij} \text{ being attained at least twice whenever } |N_i| = 3.$$

*Proof.* Fix  $m \in [4]$  and write  $Q_m(u, v) := q(e_m + e_u + e_v)$ . For distinct  $j, k \neq m$ ,

$$Q_m(m, m) = -r_m, \quad Q_m(m, j) = -s_{mj}, \quad Q_m(j, j) = -s_{jm}, \quad Q_m(j, k) = 0.$$

Indeed,  $e_m + e_j + e_k = z_\ell$  for the remaining index  $\ell$ . Let  $j, k, \ell$  be the three indices distinct from  $m$ . Pairing  $q$  with the elementary directions corresponding to  $(2e_j, e_k + e_\ell)$ ,  $(2e_j, e_m + e_k)$ , and  $(2e_m, e_j + e_k)$  gives (A.7), (A.8), and (A.9). If  $N_m = \{j, k\}$ , then  $e_m + e_\ell$  is absent from  $S_m$ , while the other four points in the corresponding rectangle are present. The degenerate four-point equality (A.4) gives  $s_{mj} = s_{mk}$ , proving (A.10).

It remains to identify the local  $M$ -concavity conditions. Every local exchange in  $\Delta_4^3$  lies in a quadratic slice: if  $\alpha, \beta \in \Delta_4^3$  and  $\|\alpha - \beta\|_1 = 4$ , then  $|\alpha \wedge \beta| = (|\alpha| + |\beta| - \|\alpha - \beta\|_1)/2 = 1$ , so  $\alpha \wedge \beta = e_m$  for some  $m$ . The diagonal–off-diagonal exchanges give (A.7)–(A.9). If  $j, k \neq m$ , the diagonal–diagonal exchange between  $2e_j$  and  $2e_k$  gives  $-s_{jm} - s_{km} \leq 0$ , which follows from (A.7). The only remaining diagonal–diagonal inequality is  $2s_{mj} \leq r_m + s_{jm}$ , coming from  $2e_m$  and  $2e_j$ . If this pair occurs, then  $p_m \in S$ , so  $|N_m| = 3$  by Lemma A.4; choosing  $k \in N_m \setminus \{j\}$  and adding  $r_m \geq s_{mj} + s_{mk}$  and  $s_{mj} \leq s_{mk} + s_{jm}$  proves it.

Only exchanges between disjoint off-diagonal pairs remain. If  $|N_m| = 2$ , the unique feasible branch gives (A.10). If  $|N_m| = 3$ , the three opposite pairings have weights  $-s_{mj}, -s_{mk}, -s_{m\ell}$ , so the exchange condition is exactly (A.11). This exhausts the local exchanges.  $\square$

The preceding lemma isolates the only branching obstruction: a function in  $\text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ$  need not itself satisfy (A.11). The next lemma removes this obstruction.

LEMMA A.6 (Star decomposition). *Let  $q : S \rightarrow \mathbb{R}$  be normalized by  $q(z_i) = 0$  and suppose that its coordinates satisfy (A.7)–(A.10). Then*

$$q = \sum_{\rho} \lambda_{\rho} q^{(\rho)}, \quad \lambda_{\rho} \geq 0,$$

for  $M$ -concave functions  $q^{(\rho)} : S \rightarrow \mathbb{R}$ .

*Proof.* For each  $i$ , define

$$m_i := \begin{cases} 0, & N_i = \emptyset, \\ s_{ij} (= s_{ik}), & N_i = \{j, k\}, \\ \min_{j \in N_i} s_{ij}, & |N_i| = 3, \end{cases} \quad u_{ij} := s_{ij} - m_i.$$

Then  $u_{ij} \geq 0$ , and

$$(A.12) \quad u_{ij} > 0 \implies |N_i| = 3.$$

Indeed, when  $|N_i| = 2$ , the two entries in row  $i$  are equal by (A.10).

For a mutual pair  $i \leftrightarrow j$ , set  $e_{ij} := \min\{u_{ij}, u_{ji}\} = e_{ji}$  and  $y_{ij} := (u_{ij} - u_{ji})_+$ ; for a non-mutual ordered pair set  $y_{ij} := 0$ . The key estimate is

$$(A.13) \quad y_{ij} \leq m_j.$$

Indeed, if  $y_{ij} > 0$ , then  $u_{ij} > 0$ , so  $|N_i| = 3$  by (A.12). Since  $j$  is not a minimizer in row  $i$ , there is  $k \neq i, j$  with  $u_{ik} = 0$ . Using (A.8),

$$m_i + u_{ij} = s_{ij} \leq s_{ik} + s_{ji} = m_i + m_j + u_{ji},$$

which proves (A.13).

We now decompose the  $s$ -coordinates. An atom below has  $q(z_i) = 0$  and all unspecified  $s$ - and  $r$ -coordinates equal to zero. For fixed  $j$ , let  $I_j := \{i \neq j : i \leftrightarrow j, y_{ij} > 0\}$ . If  $m_j = 0$ , then  $I_j = \emptyset$  by (A.13). Assume  $m_j > 0$  and put  $\theta_{ij} := y_{ij}/m_j$ . For  $J \subseteq I_j$ , set

$$\pi_{j,J} := \prod_{i \in J} \theta_{ij} \prod_{i \in I_j \setminus J} (1 - \theta_{ij}).$$

Then  $\sum_J \pi_{j,J} = 1$  and  $\sum_{J \ni i} \pi_{j,J} = \theta_{ij}$ .

Define the star atom  $H_{j,J}$  by  $s_{jk} = 1$  for  $k \in N_j$  and  $s_{ij} = 1$  for  $i \in J$ , together with  $r_j = 2$  when  $p_j \in S$  and  $r_i = 1$  when  $i \in J$  and  $p_i \in S$ . The sum  $\sum_{J \subseteq I_j} m_j \pi_{j,J} H_{j,J}$  contributes  $m_j$  to every outgoing edge from  $j$  and  $y_{ij}$  to every asymmetric incoming edge  $i \rightarrow j$ .

After removing all star atoms, a mutual pair  $i \leftrightarrow j$  has the same residual  $e_{ij}$  in both directions. Remove it with the two-cycle atom  $C_{ij}$  defined by  $s_{ij} = s_{ji} = 1$  and, when present,  $r_i = r_j = 1$ . If  $i \rightarrow j$  is non-mutual, remove its residual  $u_{ij}$  with the one-edge atom  $E_{ij}$  defined by  $s_{ij} = 1$  and, when present,  $r_i = 1$ . All  $s$ -coordinates are now matched.

Suppose  $p_i \in S$ . Then  $|N_i| = 3$  and  $\min_{j \in N_i} u_{ij} = 0$ . By (A.9),

$$r_i \geq \max_{\substack{j, k \in N_i \\ j \neq k}} (s_{ij} + s_{ik}) = 2m_i + \sum_{j \in N_i} u_{ij}.$$

The atoms already chosen contribute exactly the right-hand side. The remaining slack  $\delta_i := r_i - 2m_i - \sum_{j \in N_i} u_{ij} \geq 0$  is removed by the pure  $r_i$  atom

whose only nonzero coordinate is  $r_i = 1$ . This gives an exact nonnegative decomposition of  $q$ .

It remains to check  $M$ -concavity. By Lemma A.5, it is enough to verify (A.7)–(A.11). The nonzero rows of the atoms are  $(1, 1)$ ,  $(1, 1, 1)$ , or  $(1, 0, 0)$ , so (A.7), (A.10), and (A.11) are immediate. The chosen  $r$ -coordinates give (A.9). For (A.8), if the left-hand side is 1, then either another center edge or the reverse edge contributes 1 on the right. For a one-edge atom the reverse edge is absent, so no such support-defined instance occurs. Thus every atom is  $M$ -concave.  $\square$

### Proof of the local-to-global theorem.

*Proof of Theorem 3.9.* The second inclusion in (A.6) gives  $\text{Cone}(\mathcal{G}_4(S)) \subseteq \text{BR}(\mathcal{L}_S^+)$ . Let  $w = (w_\alpha)_{\alpha \in S} \in \text{BR}(\mathcal{L}_S^+)$ . We first record

$$(A.14) \quad \sum_{\alpha \in S} w_\alpha \alpha = 0.$$

Indeed,  $q \equiv 0$  in (A.1) shows that  $\mathcal{L}_S^+$  is nonempty. For  $a \in \mathbb{R}^4$  and  $t \in \mathbb{R}$ , the change of variables  $x_i \mapsto e^{ta_i} x_i$  preserves  $\mathcal{L}_S^+$  and multiplies  $R_w$  by

$$\exp \left( t \left\langle a, \sum_{\alpha \in S} w_\alpha \alpha \right\rangle \right).$$

Boundedness for all  $t$  gives (A.14).

Suppose first that no slice is branching. If  $w \notin \text{Cone}(\mathcal{G}_4^{\text{el}}(S))$ , separation gives  $q \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ$  with  $\langle w, q \rangle > 0$ . Every local exchange lies in a quadratic slice. The only quadratic exchange with two different branches is the exchange between two disjoint off-diagonal pairs; if both branches were present, the slice would be branching. Hence every local exchange has a unique feasible branch, and its inequality is one of (A.2), (A.3), or (A.4). Thus  $q$  is  $M$ -concave. The family (A.1) lies in  $\mathcal{L}_S^+$ , while  $R_w(f_t) = e^{t\langle w, q \rangle} \rightarrow \infty$ , a contradiction. Therefore  $w \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S)) \subseteq \text{Cone}(\mathcal{G}_4(S))$ .

Now suppose that some slice  $S_i$  is branching. By Lemma A.2, either  $S = e_i + T$  or  $z_1, z_2, z_3, z_4 \in S$ . In the first case, Lemma A.3 gives  $\text{BR}(\mathcal{L}_S^+) = \text{BR}(\mathcal{L}_T^+)$ . Since  $S_i = T$ , under the natural identification of coefficient coordinates  $w$  is the extension of an element of  $\text{BR}(\mathcal{L}_{S_i}^+)$ , so  $w \in \mathcal{G}_4(S) \subseteq \text{Cone}(\mathcal{G}_4(S))$ .

It remains to treat  $z_1, z_2, z_3, z_4 \in S$ . Suppose  $w \notin \text{Cone}(\mathcal{G}_4^{\text{el}}(S))$ . Choose  $q \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ$  with  $\langle w, q \rangle > 0$ . Let  $\ell(\alpha) = \langle b, \alpha \rangle$  be the unique linear function for which  $q' := q - \ell$  satisfies  $q'(z_i) = 0$  for all  $i$ . By (A.5),  $\langle h, \ell \rangle = 0$  for every  $h \in \mathcal{G}_4^{\text{el}}(S)$ , so  $q' \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S))^\circ$ . By (A.14),  $\langle w, \ell \rangle = 0$ , hence  $\langle w, q' \rangle > 0$ .

Lemma A.5 gives (A.7)–(A.10) for  $q'$ , and Lemma A.6 gives  $q' = \sum_{\rho} \lambda_{\rho} q^{(\rho)}$  with  $\lambda_{\rho} \geq 0$  and every  $q^{(\rho)}$   $M$ -concave. Since  $\langle w, q' \rangle > 0$ , some summand has  $\langle w, q^{(\rho)} \rangle > 0$ . Applying (A.1) to this summand gives a family in  $\mathcal{L}_S^+$  along which  $R_w$  tends to infinity, a contradiction. Thus  $w \in \text{Cone}(\mathcal{G}_4^{\text{el}}(S)) \subseteq \text{Cone}(\mathcal{G}_4(S))$ .

Hence  $\text{BR}(\mathcal{L}_S^+) \subseteq \text{Cone}(\mathcal{G}_4(S))$ . Together with (A.6), this proves the theorem.  $\square$

*Remark A.7* (Where the cubic hypothesis enters). The degree-three hypothesis enters in the branching analysis. Since  $\Delta_4^{3-2} = \{e_1, e_2, e_3, e_4\}$ , every quadratic slice is indexed by a single coordinate. Lemma A.2 then uses  $|\alpha| = 3$  to reduce a branching support either to a translated quadratic support or to the four points  $z_1, \dots, z_4$ . Once all four  $z_i$  are present, every point of  $\Delta_4^3$  is one of the  $z_i$ ,  $a_{ij}$ , or  $p_i$ , which makes the support-graph decomposition possible.

A.2. *A quaternary quartic counterexample.* We now show that the preceding theorem does not extend to degree four.

Let  $S = \Delta_4^4$ . For  $\beta \in \Delta_4^2$  and pairwise distinct  $i, j, k \in [4]$ , set

$$T_{\beta}^{ij|k}(f) := \frac{c_{\beta+e_i+e_j} c_{\beta+2e_k}}{c_{\beta+e_i+e_k} c_{\beta+e_j+e_k}},$$

and let  $g_{\beta}^{ij|k}$  be its exponent vector. By Lemma A.1,  $T_{\beta}^{ij|k} \leq 2$ . Every quadratic slice now has full support, and Huang–Huh–Soskin–Wang show that the bounded-ratio cone of a positive full-support  $4 \times 4$  Lorentzian matrix is generated by the triangular ratios [5, Theorem B]. Therefore

$$(A.15) \quad \text{Cone}(\mathcal{G}_4(S)) = \text{Cone} \left\{ g_{\beta}^{ij|k} : \beta \in \Delta_4^2, i, j, k \in [4] \text{ pairwise distinct} \right\}.$$

We abbreviate  $c_{(a,b,c,d)}$  by  $c_{abcd}$ .

For convenience, we restate Theorem 3.10 before giving its proof.

THEOREM 3.10 (Quaternary quartics). *For  $S = \Delta_4^4$ ,*

$$\text{Cone}(\mathcal{G}_4(S)) \subsetneq \text{BR}(\mathcal{L}_S^+).$$

*More precisely,*

$$(3.1) \quad R(f) := \frac{c_{1030} c_{3010} c_{0202}}{c_{2020} c_{1012} c_{1210}}$$

*satisfies  $R(f) \leq 64$  for every  $f \in \mathcal{L}_S^+$ , but its exponent vector does not belong to  $\text{Cone}(\mathcal{G}_4(S))$ . The constant 64 is not claimed to be optimal.*

*Proof.* Let  $w_R$  be the exponent vector of  $R$ . We first prove that  $R$  is bounded. Fix  $\beta_0 = (0, 1, 0, 1)$  and write  $H_{\beta_0}(f) = (p_{ij})$ . Set

$$A := c_{1201} c_{0112} = p_{12} p_{34}, \quad B := c_{1111} c_{0202} = p_{13} p_{24},$$

$$C := c_{1102} c_{0211} = p_{14} p_{23}.$$

Lemma A.1 gives  $\sqrt{B} \leq \sqrt{A} + \sqrt{C}$ , and therefore

$$(A.16) \quad \min \left\{ \frac{B}{A}, \frac{B}{C} \right\} \leq 4.$$

Indeed, otherwise both  $\sqrt{B/A}$  and  $\sqrt{B/C}$  would exceed 2. Direct cancellation gives

$$(A.17) \quad R = T_{1010}^{24|3} T_{1010}^{43|1} T_{1100}^{24|3} T_{0011}^{24|1} \frac{B}{A},$$

$$(A.18) \quad R = T_{1010}^{24|3} T_{1010}^{23|1} T_{0110}^{24|1} T_{1001}^{24|3} \frac{B}{C}.$$

Each triangular factor is at most 2. Choosing the factorization corresponding to the smaller term in (A.16) gives  $R \leq 2^4 \cdot 4 = 64$ , so  $w_R \in \text{BR}(\mathcal{L}_S^+)$ .

We now separate  $w_R$  from the local cone. Place the four indices at the vertices of a square,  $1 = (0, 0)$ ,  $2 = (1, 0)$ ,  $3 = (1, 1)$ ,  $4 = (0, 1)$ , and let  $\rho$  be the graph distance. For  $\alpha \in \Delta_4^4$ , define

$$(A.19) \quad q_\alpha^0 := \min_{r \in [4]} \sum_{i=1}^4 \alpha_i \rho(i, r).$$

Thus  $q_\alpha^0$  is the minimum total number of edge-moves required to move the four tokens described by  $\alpha$  to one common vertex. We claim that  $\langle g_\beta^{ij|k}, q^0 \rangle \leq 0$  for every triangular generator. Let  $P_1 = \{1, 4\} \mid \{2, 3\}$  and  $P_2 = \{1, 2\} \mid \{3, 4\}$  be the two coordinate cuts of the square, and put  $\mu(s) := \min\{s, 4 - s\}$ , so  $(\mu(0), \dots, \mu(4)) = (0, 1, 2, 1, 0)$ . For a cut  $P = A \mid A^c$ , set  $q_\alpha^P := \mu(\sum_{\ell \in A} \alpha_\ell)$ . The square metric is the sum of the two cut metrics. Moreover, the minimizing side for each cut may be chosen independently, since the intersection of any two chosen sides is a vertex of the square. Hence  $q_\alpha^0 = q_\alpha^{P_1} + q_\alpha^{P_2}$ .

It is therefore enough to check one cut. Write  $b := \sum_{\ell \in A} \beta_\ell \in \{0, 1, 2\}$  and let  $\varepsilon_r = 1$  if  $r \in A$  and 0 otherwise. The required inequality is

$$(A.20) \quad \mu(b + \varepsilon_i + \varepsilon_j) + \mu(b + 2\varepsilon_k) \leq \mu(b + \varepsilon_i + \varepsilon_k) + \mu(b + \varepsilon_j + \varepsilon_k).$$

The three distinct vertices  $i, j, k$  are split by the cut as  $2 + 1$ . If  $k$  lies with exactly one of  $i, j$ , the two sides of (A.20) are equal. Otherwise  $i, j$  lie on one side and  $k$  on the other; after interchanging the sides, the inequality becomes  $\mu(b) + \mu(b + 2) \leq 2\mu(b + 1)$  for  $b = 0, 1, 2$ , which follows from the discrete concavity of  $0, 1, 2, 1, 0$ . Thus

$$(A.21) \quad \langle g_\beta^{ij|k}, q^0 \rangle \leq 0 \quad \text{for every } g_\beta^{ij|k}.$$

Modify one coordinate by setting  $q_{2020} := q_{2020}^0 - 2 = 2$  and  $q_\alpha := q_\alpha^0$  for  $\alpha \neq (2, 0, 2, 0)$ . We still have

$$(A.22) \quad \langle g_\beta^{ij|k}, q \rangle \leq 0 \quad \text{for every } g_\beta^{ij|k}.$$

Indeed, if  $\varepsilon_{2020}$  occurs with positive sign, lowering  $q_{2020}$  only decreases the pairing. The only generators in which it occurs with negative sign are

$$g_{1010}^{12|3}, \quad g_{1010}^{14|3}, \quad g_{1010}^{23|1}, \quad g_{1010}^{34|1}.$$

The symmetries of the square preserving the diagonal  $\{1, 3\}$  also preserve  $q$  and act transitively on these four generators. For the first,

$$\langle g_{1010}^{12|3}, q \rangle = q_{2110} + q_{1030} - q_{2020} - q_{1120} = 3 + 2 - 2 - 3 = 0.$$

This proves (A.22).

Finally, the  $q^0$ -values on the three numerator coefficients in (3.1) are 2, 2, 4, while those on the three denominator coefficients are 4, 2, 2. Both sums are 8, so  $\langle w_R, q^0 \rangle = 0$ . Since the exponent of  $c_{2020}$  in  $R$  is  $-1$  and  $q_{2020} = q_{2020}^0 - 2$ , we obtain  $\langle w_R, q \rangle = 2 > 0$ . By (A.15) and (A.22),  $q$  is nonpositive on  $\text{Cone}(\mathcal{G}_4(S))$ , whereas it is positive on  $w_R$ . Hence  $w_R \notin \text{Cone}(\mathcal{G}_4(S))$ .  $\square$

*Remark A.8.* The two subsections give a sharp degree boundary in four variables: the local-to-global theorem holds for cubics with arbitrary  $M$ -convex support and fails for quartics even at full support. We have also found a full-support cubic counterexample in five variables, showing that the cubic result is sharp in the number of variables as well. We do not pursue this direction here.

## Appendix B. Bounded ratios and median-graph transportation

The transportation cost used in the proof of Theorem 3.10 admits a general interpretation. We show that transportation costs on finite median graphs give polar directions for full-support bounded-ratio cones in arbitrary dimension and degree.

Throughout this appendix, let  $n \geq 2$  and  $d \geq 1$ , and write

$$\mathcal{L}_{n,d}^+ := \mathcal{L}_{\Delta_n^d}^+.$$

Let  $P = A \mid A^c$  be a cut of  $[n]$ . For  $\alpha \in \Delta_n^d$ , write

$$\alpha(A) := \sum_{i \in A} \alpha_i,$$

and define

$$q_\alpha^P := \min\{\alpha(A), d - \alpha(A)\},$$

the minimum number of tokens that must cross the cut in order to move all tokens to one side. This is independent of the choice of side  $A$  of the cut  $P$ .

LEMMA B.1 (Cut directions). *For every cut  $P$  of  $[n]$ ,*

$$q^P \in \text{BR}(\mathcal{L}_{n,d}^+)^{\circ}.$$

Equivalently,

$$\langle w, q^P \rangle \leq 0 \quad \text{for every } w \in \text{BR}(\mathcal{L}_{n,d}^+).$$

*Proof.* Write

$$\mu_d(s) := \min\{s, d - s\},$$

so that  $q_\alpha^P = \mu_d(\alpha(A))$ . We first show that  $q^P$  is  $M$ -concave. Let  $\alpha, \gamma \in \Delta_n^d$  and suppose that  $\alpha_i > \gamma_i$ .

Assume first that  $i \in A$ . If  $\alpha(A) \leq \gamma(A)$ , there is  $j \in A$  with  $\alpha_j < \gamma_j$ , and the corresponding exchange leaves both  $A$ -masses unchanged. If  $\alpha(A) > \gamma(A)$ , there is  $j \in A^c$  with  $\alpha_j < \gamma_j$ . Writing  $a = \alpha(A)$  and  $b = \gamma(A)$ , we have  $a > b$ , and the discrete concavity of  $\mu_d$  gives

$$\mu_d(a) + \mu_d(b) \leq \mu_d(a - 1) + \mu_d(b + 1).$$

This is precisely the required exchange inequality. The case  $i \in A^c$  follows by interchanging the two sides of  $P$ . Thus  $q^P$  is  $M$ -concave.

By [4, Theorem 3.14], for every  $t \geq 0$ ,

$$f_{P,t}(\mathbf{x}) := \sum_{\alpha \in \Delta_n^d} e^{tq_\alpha^P} \frac{\mathbf{x}^\alpha}{\alpha!}$$

belongs to  $\mathcal{L}_{n,d}^+$ . If  $w \in \text{BR}(\mathcal{L}_{n,d}^+)$ , then  $R_w$  is uniformly bounded on  $\mathcal{L}_{n,d}^+$ , while

$$R_w(f_{P,t}) = e^{t\langle w, q^P \rangle}.$$

Letting  $t \rightarrow \infty$  gives  $\langle w, q^P \rangle \leq 0$ .  $\square$

We now recall the standard cut structure of finite median graphs [3]. Let  $G$  be a finite median graph with vertex set  $[n]$ . Its edge  $\Theta$ -classes determine convex cuts  $P_r = H_r \mid H_r^c$ ,  $1 \leq r \leq m$ , such that

$$(B.1) \quad d_G(u, v) = \#\{r : P_r \text{ separates } u \text{ and } v\}.$$

Moreover, convex subsets of a finite median graph satisfy the Helly property.

For  $\alpha \in \Delta_n^d$ , define the transportation cost

$$(B.2) \quad q_\alpha^G := \min_{v \in [n]} \sum_{i=1}^n \alpha_i d_G(i, v).$$

In fact, this is the minimum transportation cost to move the  $d$  tokens described by  $\alpha$  to one common vertex.

**THEOREM B.2** (Median-graph transportation directions). *Let  $G$  be a finite median graph on  $[n]$ . Then*

$$q^G = \sum_{r=1}^m q^{P_r},$$

and consequently

$$q^G \in \text{BR}(\mathcal{L}_{n,d}^+)^{\circ}.$$

Equivalently, every  $w \in \text{BR}(\mathcal{L}_{n,d}^+)$  satisfies

$$\langle w, q^G \rangle \leq 0.$$

*Proof.* Fix  $\alpha \in \Delta_n^d$ . By (B.1), for every  $v \in [n]$ ,

$$\sum_{i=1}^n \alpha_i d_G(i, v) \geq \sum_{r=1}^m \min\{\alpha(H_r), \alpha(H_r^c)\}.$$

For each  $r$  with unequal masses, let  $M_r$  be the side of  $P_r$  having mass greater than  $d/2$ . Any two such sides intersect, since otherwise their masses would sum to more than  $d$ . The sets  $M_r$  are convex, so the Helly property gives

$$\bigcap_r M_r \neq \emptyset.$$

Choose  $v$  in this intersection. For every cut with unequal masses,  $v$  lies on the majority side, so exactly the minority mass crosses that cut. For a cut with equal masses, either side gives the same contribution. Hence

$$q_{\alpha}^G = \sum_{r=1}^m \min\{\alpha(H_r), \alpha(H_r^c)\} = \sum_{r=1}^m q_{\alpha}^{P_r}.$$

Thus  $q^G = \sum_r q^{P_r}$ . By Lemma B.1, each  $q^{P_r}$  belongs to  $\text{BR}(\mathcal{L}_{n,d}^+)^{\circ}$ , and therefore so does  $q^G$ .  $\square$

*Remark B.3.* For the square  $G = C_4$  with the labeling used in (A.19), the two  $\Theta$ -classes are exactly the two coordinate cuts used in the proof of Theorem 3.10. Thus Theorem B.2 shows that the unmodified transportation direction  $q^0$  is nonpositive on the entire bounded-ratio cone  $\text{BR}(\mathcal{L}_{4,4}^+)$ , rather than only on the local cone in (A.15).

More generally, taking  $G$  to be the hypercube  $Q_k$  gives the Hamming transportation direction in  $2^k$  variables and every degree.

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